

Transfer Report

Mathematical and Computational Modelling of the Intra-Cellular Dynamics
Present in a Pneumonic Plague Infection

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1 Introduction

I am a PhD student with an EPSRC CASE studentship, in collaboration with the Defence Science and Technology Laboratory (DSTL), a government organisation that carries out science and technology research in the defence and security field. Using a combination of mathematical modelling and computational simulation, I hope to uncover a better understanding of *Yersinia pestis* and plague pathogenesis. The end goal being to create a computational toolset that can aid in the assessment of exposure risks when working with the plague bacteria in a lab.

This report will cover two mathematical approaches to modelling dynamics of plague bacteria inside a single cell. The first uses a Markov process that is analysed with first step analysis. Then the second works with a similarly constructed model in continuous time and uses an approach with coupled ordinary differential equations. Both methods follow a theme of trying to calculate the likelihood that a plague infected cell will ultimately survive.

Computational realisations with the Gillespie algorithm simulate the dynamics and are used as a means of verification for analytic results.

2 Literature Review

2.1 History of the Plague *Bacillus* and its Effect on Human Societies

Gram-negative bacterium, *Y. pestis*, named after Alexandre Yersin who discovered it in 1894, is the etiological agent of plague. The term Gram-negative comes from a bacterium's propensity to be incompatible with a method of analysis that involves staining samples with a violet dye. Various types of bacteria have this trait due to a thin cell wall composed of a substance called peptidoglycan [2].

There have been a few major pandemics of the plague in recorded history, namely, Justinian's plague 541-767AD, The Black Death 1347-1351, and the Great Plague 1665-1667. Interestingly, during the latter, it was said that these 18 months were used by Newton to lay the foundations for his famous discoveries in mathematics and physical science. In addition to these, there was a global outbreak during the second half of the 19th century, creeping into the beginnings of the 20th century in some areas.

Despite the fact that these great tragedies did occur is indisputable, the association of each of these pandemics with, specifically, *Yersinia pestis* remains controversial, partly because the history of this largely monomorphic bacterium was unknown [1]. It is relevant to consider that the *Yersinia* genus as a whole contains three main sub-species: *Y. pestis*, *Y. pseudotuberculosis*, and *Y. enterocolitica*. Although *Y. pestis* differs from the other two in that it is spread by fleas that are usually resident in rodents, whereas *pseudotuberculosis* and *enterocolitica* are enteric food and waterborne pathogens. Current thinking is of the conviction that *Y. pestis* evolved from *Y. pseudotuberculosis* 1,500 - 20,000 years ago [3] & [4]. As well as a distinction of sub-species of the *Yersinia* genus, there is also a documentation regarding three distinct biovars of *Yersinia pestis*, each with subtle phenotypic differences.

2.2 Mechanisms of Pathogenesis

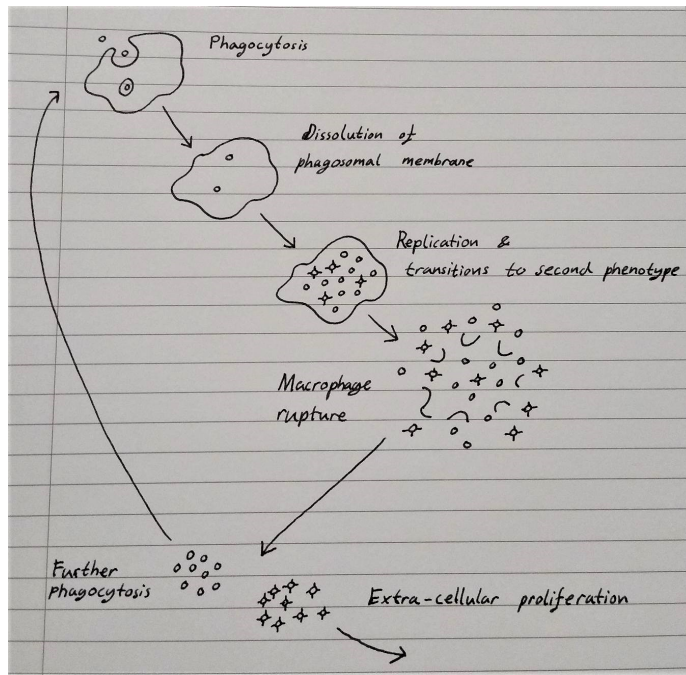
Depending on where the infection is localised in the body, plague can be divided into three distinct types: bubonic, septicaemic, and pneumonic. Bubonic plague comes about when the bacteria takes over your lymph nodes forming enlarged buboes; septicaemic plague is characterised by an infection taking hold of blood in the circulatory system, and can come about either independently or in the later stages of one of the other two versions of the disease; and pneumonic plague, the one we are focusing our attention on, is centred in the lungs.

Unlike bubonic plague, which is generally a flea borne disease, at the outset of a pneumonic plague infection, *Yersinia pestis* is taken into the body through alveoli in the lung after the organism breathes in its bacterial particles. There the bacteria is susceptible to absorption by a couple of key types of immune cell, macrophages and neutrophils, which belong to the class called leukocytes – these are also known as white blood cells. This process of absorption is called phagocytosis. “*Y. pestis* is typically killed in neutrophils, whereas in macrophages, it can survive and acquire antiphagocytic capabilities, which enable its extracellular survival *in vivo*” [5]. At least, this has been, and to an extent lingers as, the prevailing view in the field; new

findings suggest that in some circumstances, plague bacteria can also survive in neutrophils. For the purposes of modelling, we will refer to the bacteria that have antiphagocytically armed themselves in a macrophage as *Y. pestis* type *b*; type *a* being the more vulnerable preceding phenotype.

While resident in the intracellular medium of a macrophage, in addition to the arming process, the bacteria can practice division, multiplying in numbers. Then at some time, their relatively safe cellular refuge disintegrates; the host macrophage ruptures via apoptosis.

Once released into the more chaotic and testing extracellular environment, of which abrasive factors include “low PH, shear stress, and host defensive mechanisms, such as recognition and killing by [...] cytotoxic T-cells”, the free pathogens can proceed with their infection process: armed type *b* bacteria can resist further phagocytosis, dividing and proliferating to different parts of the body, such as the lymph nodes; particles that are not yet armed (type *a*) are susceptible to further phagocytosis by macrophages.

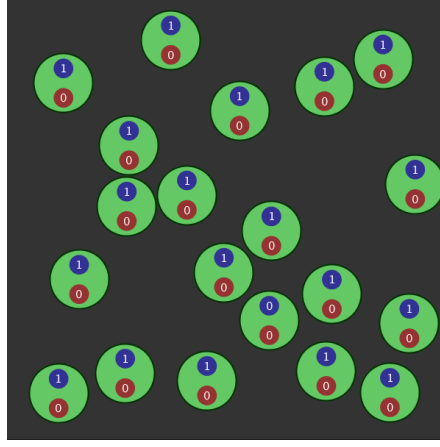


Yersinia pestis inter-macrophage dynamics (type *b* *Y. pestis* shown as crossed circles; type *a*, regular circles).

It is worth noting that there are a handful of intricacies here; type *b*, and *a* for that matter, are also able to enter host cells actively through a process called zippering where they present ligands on their surface which bind to immune cells; in addition to the antiphagocytic property of type *b*, *Y. pestis* also releases a variety of Yop proteins (YopE, YopH, YopM, YopO/YpkA YopP/J, and YopT) via a type III secretion system (T3SS), these then inhibit phagocytosis and block pro-inflammatory signals. So it appears that as the infection develops, phagocytes become unable to phagocytose in general – both type *a* and *b* bacteria. This is supported by the fact that autopsies of plague victims show a great deal of extracellular bacteria, but little evidence of phagocytosis [5]. [6] also talks about how a pneumonic plague infection is biphasal. That is, it happens in two stages: first the bacteria gets a foothold utilising the intra-macrophage refuge, then, once better established, spreads more freely in the extracellular environment. For the purpose of getting a model up and running, we will leave these further intricacies aside, and lay out our assumptions thus.

3 Rules of the game

Initially, there are a number (M) of macrophages floating around the extracellular environment. $m \leq M$ of these contain from the outset a single type *a* *Y. pestis* particle; the rest contain neither type *a* or *b* bacteria. As the system progresses in continuous time, a range of events can occur:



Snapshot taken from computational model before any event occurrences ($M = 20, m = 19$).

- type a & b Y. pestis bacteria can divide within the host macrophage,
- type a , through a process of “arming” can differentiate into type b Y. pestis,
- bacterial particles of each type can die, naturally or as a result of the macrophage’s immuno-toxins,
- and the macrophages can rupture, releasing their pathogenic load into the extracellular environment.

Each of the possible events has an associated rate defining the likelihood of it happening in a given time period. Notation for these is prescribed as follows:

λ_a – Birth rate of initial type of Y. Pestis (type a)

μ_a – Death rate for type a

λ_b – Birth rate of second phenotype (type b)

μ_b – Death rate for type b

γ – Transition rate from type a to b

δ – Macrophage rupture event rate

All with the exception of the rupture rate δ relate to the dynamics of a single cell. Extending beyond this, we must also consider how the probability of an event, say a division, changes as there become multiple present bacteria of differing type; how does the probability that a new bacterial particle will be birthed within a given macrophage change as the number of potential parent bacteria increases? One may simply prescribe a linear relationship here: birth rate $\lambda_a^{(n)} = \lambda_a n_a$ with n_a type a bacteria present. This works to keep the complication of the subsequent analysis in check, and for now we will propose the model as such. For birth rates in particular, there are non-linear relationships that can be explored that may for example take into account the fact of limited resources (see below), but when it comes to Y. pestis death events, the linear dynamics seem like a clear choice: $\mu_i^{(n)} = \mu^{(n)} n_i$ for $i = a, b$. When it comes to macrophage rupture events, the question is one of whether the number of resident bacteria brings forward the inevitable apoptosis, or if the leukocyte simply goes about its life cycle unperturbed by its stowaway passengers. It’s also worth considering whether the age of a macrophage should play a key role in its chance of rupture at a given instance. In any case, for now, we can keep things simple choosing

$$\delta_n = \delta n$$

where $n = n_a + n_b$ is the total number of bacteria hosted.

In the following mathematical analysis, there will be a cap, N , imposed on the number of *Y. pestis* bacteria that can inhabit a macrophage. This is defined as

$$n_a + n_b = n \leq N.$$

Before getting to this though, let us consider a few potential avenues for future exploration in the rules and dynamics:

1.) We have chosen the birth rate scaling to be linear with respect to the number of bacteria present, but this can risk missing out on some fundamental dynamics. For example, it may make more sense to prescribe a logistic relationship between the birth rate and number of present bacteria,

$$\lambda_a^{(n)} = \lambda_a \left(1 - \frac{n_a + n_b}{\kappa} \right),$$

where n_a and n_b are the respective numbers of the different types of *Y. pestis* that are present, and κ is some carrying capacity of the macrophage, as this can capture the influence of the availability of intracellular resources necessary for division. The carrying capacity, κ , may be chosen to be N , the maximum permitted number of bacteria; it may be chosen to be less than κ , which may result in a fluctuation in bacterial count that is partially restricted by a ceiling, provided κ is sufficiently close to N ; or it may be fixed above N , causing only a reduction in population growth rate before the limit is reached.

2.) As we progress, it will be necessary to add the feature of re-uptake of type *a* bacteria by macrophages post apoptosis release. This can be spatially dependent or not as we see fit.

3.) We may wish to model the dynamics introduced in which secreted Yop proteins generally inhibit phagocytosis of both type *a* and *b* *Y. pestis*. When a re-uptake of type *a* bacteria by macrophages is introduced to the model, this would come in as a modification to the associated rate of that re-uptake which is dependent of the number of extracellular bacteria, and the time they have been present (– the time they have been secreting the Yops).

4 Computational realisations of the model

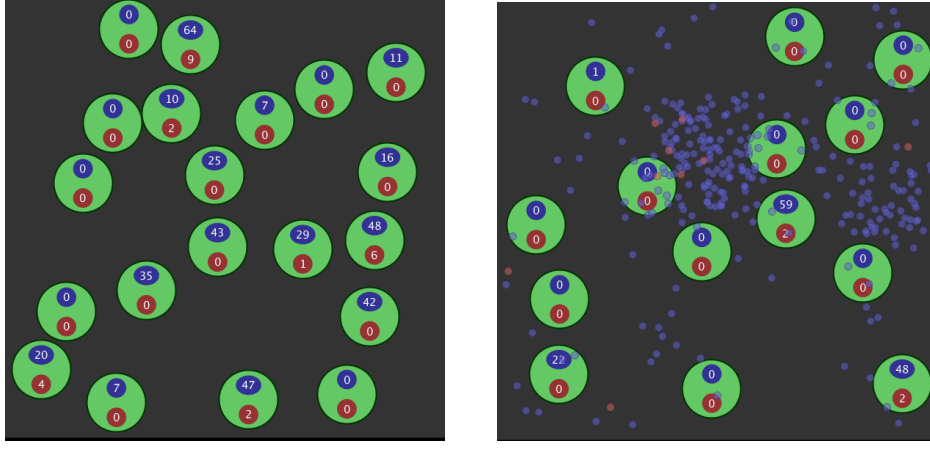
With the dynamics of the system well defined, it is possible to implement the model in a continuous-time computational form using the Gillespie algorithm. This has been done in two instances: in a framework called Processing I have written a program that produces a visual representation of the model’s realisations, I have also ported the fundamental back-end workings of this program to Python so it can be run over a large number of episodes. Part of the plan involves constructing a mathematical model that describes this system.

Using the Python implementation in this way we can calculate numerical approximations for chosen descriptors – such as probability of macrophage recovery – by observing the proportion of the many digital realisations that result in the appropriate behaviour associated with our chosen descriptor. Python is notably better suited for this purpose as it offers superior functionality in the domain of managing, manipulating, and graphing large data arrays. By this regime, we can check that the dynamics suggested by our abstracted mathematical conception match with those produced by the more concretely lucid workings of a logical machine.

Given there is an agreement between the two outputs, we can assume that the mathematical model is likely to be based on sound thinking.

4.1 Visual simulation in the Processing framework

Processing (processing.org) is a Java-based tool that was originally designed for creative coding, and specifically for use on projects that feature a significant visual element. Given this enhanced artistic capability, Processing facilitates a dynamic graphical representation of the biological system’s workings that is more lucid than a standard textual description.



In the images above you see a single realisation of the model at two instances in time. On the right we see that some of the green-circle macrophages have already ruptured, releasing their resident blue-circle type a bacteria and red-circle type b ; looking left we see the episode earlier on, though not at the beginning when macrophages hold only up to one invading bacteria. Between the first image and the second, the macrophages that have apoptosed with the ruptured event of rate δ have released into the grey extracellular environment respective numbers of type a & b *Y. pestis* shown in their interior numbered blue and red circles. (Note: in this implementation, there is currently no cap, N , on the number of bacteria in a macrophage). Click [here](#) to see a video demonstration.

Options for taking this visual implementation further are only limited by the scope of imagination, but for example could include: re-uptake of extracellular *Yersinia pestis* by macrophages through phagocytosis or other active means of entry, and this could be respecting of spatial positions and particle contact or not as desired; introduction of fresh macrophages as ruptures occur; and indeed the inclusion of other types of leucocyte, notably neutrophils as these also appear to be quite relevant in a pneumonic plague infection [7].

4.2 Tracking the dynamics numerically with Python

As was introduced at the beginning of this section, we can use a Python implementation of the model, realised over many iterations, to calculate numerical approximations that can be used to verify the following mathematical analysis. Presently, as in the subsequent mathematical discourse, we will focus our attention to the goings-on of a single macrophage – one that at initial time contains a single *Y. pestis* bacterium.

I have ported the appropriate functioning of the Processing implementation of the model across to Python. We can choose the maximum number of bacteria that a macrophage can support, N , the rate parameters, and the number of iterations to be completed. From an episode of the dynamics, when it has terminated in one of the absorption events – macrophage recovery, or rupture – the code records:

- The outcome of the process in string form, either ‘recovery’ or ‘rupture’, along with the termination time as a float.
- A time series of the numbers of bacteria n_a and n_b , as a list of integer pairs each with an associated float for the time at which that state was reached.

Then the recordings from that particular episode are stored in a list along with data from previous iterations. Accessing lists of this form from a number of experiments, I have produced the following visual representations. In figure 4 we have a number of overlayed elements: The dashed green and red vertical lines show the average time taken to reach recovery and rupture respectively, considering only the realisations that ended as such. On the same theme, the inverted stacked histogram in the lower subplot shows relatively the frequencies that each of these outcomes have occurred within consecutive time intervals. Moving to the upper subplot, we see the average number of type a bacteria, n_a ($\bar{n}_a$), over time represented by the single blue line; and the corresponding average value of n_b ($\bar{n}_b$) in the orange line.

It is worth pausing for a moment here to say how I reconciled the multiple time series data items, problematically of non-uniform time points, into a single running average. Because the Gillespie algorithm functions in the freedom of continuous-time, the time points relevant to the population changes of one iteration of the model will be different from that of another; values cannot be simply summed pointwise for the averaging process. Rather, some standardisation is necessary:

1. A uniform sequence of time steps is chosen, $\alpha, 2\alpha, 3\alpha, \dots$ for some $\alpha \in \mathbb{R}$.
2. Assign each non-uniform time point in every recorded time series to the closest higher than or equal to uniform time step using

$$T_u = \left\lceil \frac{T_c}{\alpha} \right\rceil \alpha$$

where T_c is the original continuous-time, and T_u is the resulting associated uniform time step.

3. Calculate the average values of n_a & n_b at every time step – taking for each the sum of values that have been assigned to that particular time point in step 2, then dividing appropriately by the number of allocations.

Average values of n_a & n_b with time over 100000 simulation runs (48832 ended in recovery; 51168 in rupture)

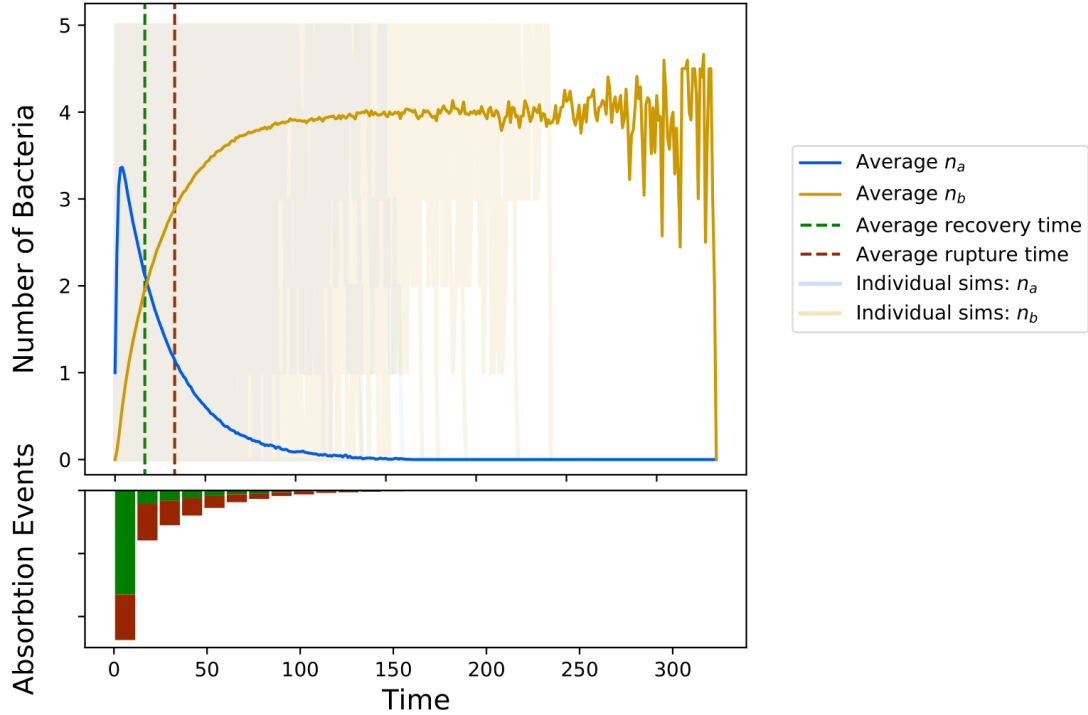


Figure 4: Created using the results from 100000 episodes of the Python implemented single-macrophage model. Upper plot shows the average numbers of bacteria, $\bar{n}_a$ & $\bar{n}_b$, over time; the lighter projections representative of values for, in this case, a sample 3000 of the individual simulations. Lower histogram displays stacked frequencies of **recovery** and **rupture** absorption events during consecutive intervals across the time scale. Parameters were set to $[N, \lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [5, 1, 0.3, 0.8, 0.3, 0.05, 0.005]$

It is clear to see in figure 4 that the average value for n_b over time, marked by the orange line, suffers from a great level of fluctuation as it progresses into the latter region. Given that the initial portion of the graph is so

smooth in comparison, one may wonder as to why this is. A little inspection, however, reveals a simple cause: by nature of the experiment, the period of time for which an episode lasts is a random variable. Necessary to this, as time progresses the number of realisations that have thus continued to be active in an epoch will decrease; the sample size for calculating $\bar{n}_a$ & $\bar{n}_b$ averages will diminish. And subsequently, the accuracy of the estimates will degenerate. A visualisation of this is seen in figure 6 (pay attention to the density of the individually plotted simulation results). In figure 4 the degeneration only occurs in $\bar{n}_b$; $\bar{n}_a$ remains smooth, sheltered from volatility by the fact that it converges to 0 prior to the “fall-off” in sample size.

Continuing on to figure 5, we see represented exactly the same data as in figure 4. Here however, the instances have been divided by outcome into one set that ended in recovery, and another of episodes ending in rupture. Post division, averages $\bar{n}_a$ & $\bar{n}_b$ were calculated and drawn, as before, using the blue and orange lines.

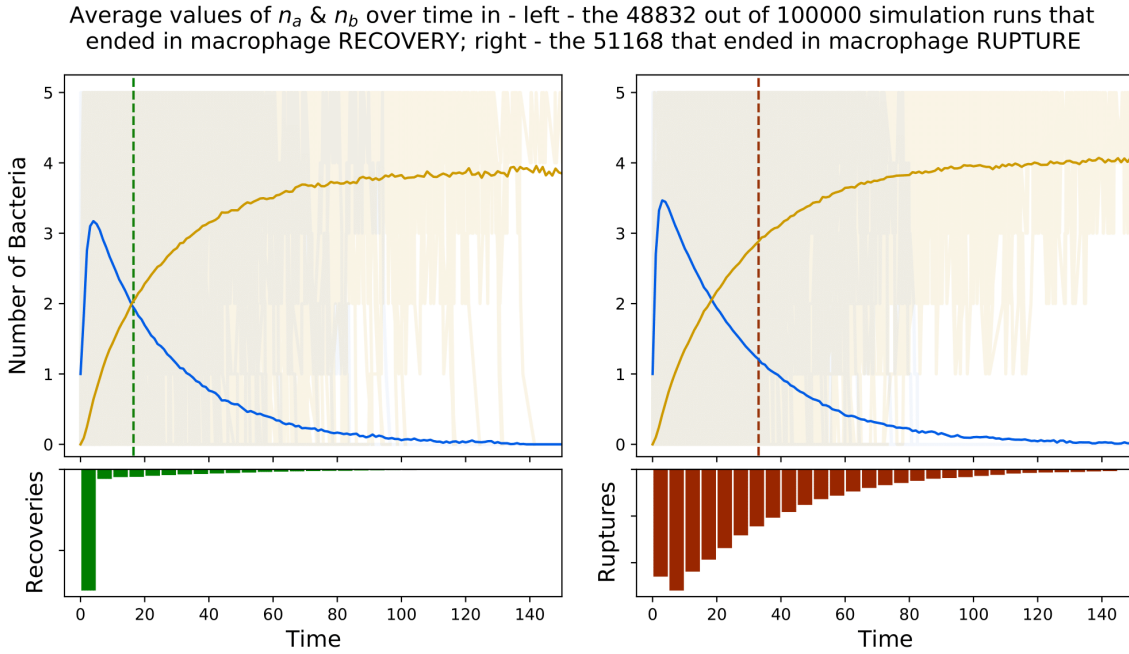


Figure 5: Data represented in this two-part figure is identical to that shown in figure 4, however, in this display the episodes represented have been divided. On the left, we see realisations of the model that ended in recovery; on the right – those that ended in rupture. In each of these, only a sample of 500 individual representations are projected.

Relevant to the parameters at hand, the following observations can be made of figure 5:

- a) On average, recovery outcomes tend to come earlier than macrophage ruptures.
- b) Not only do the recovery outcomes generally happen sooner, their distribution is also tighter than that of rupture times.
- c) $\bar{n}_a$ initially grows to a peak, then convergently declines to 0 after type b populations gets a foothold.
- d) $\bar{n}_b$, at least prior to a flattening-off around $\bar{n}_b = 4$, appears to be strictly increasing, with a rate of increase in this range that appears to be strictly decreasing.
- e) In the case of rupture outcome runs, the initial peak in the average value $\bar{n}_a$ is slightly higher.

With regard to observation a) and b), an explanation could be found in the fact that many of the recoveries likely occurred in the first simulation event. Given that the death rates are set rather smaller than the birth rates (0.3 compared to 1 & 0.8), total population size will tend to drift positively on average. With this, we

would expect that a system with two bacteria, $N = 2$, is significantly less likely to terminate in recovery than rupture – which possibly explains the sharp focus of the distribution of recovery times close to time 0.

Average values of n_a & n_b with time over all 5000 simulation runs - 1497 of which ended in recovery; 3503 in rupture

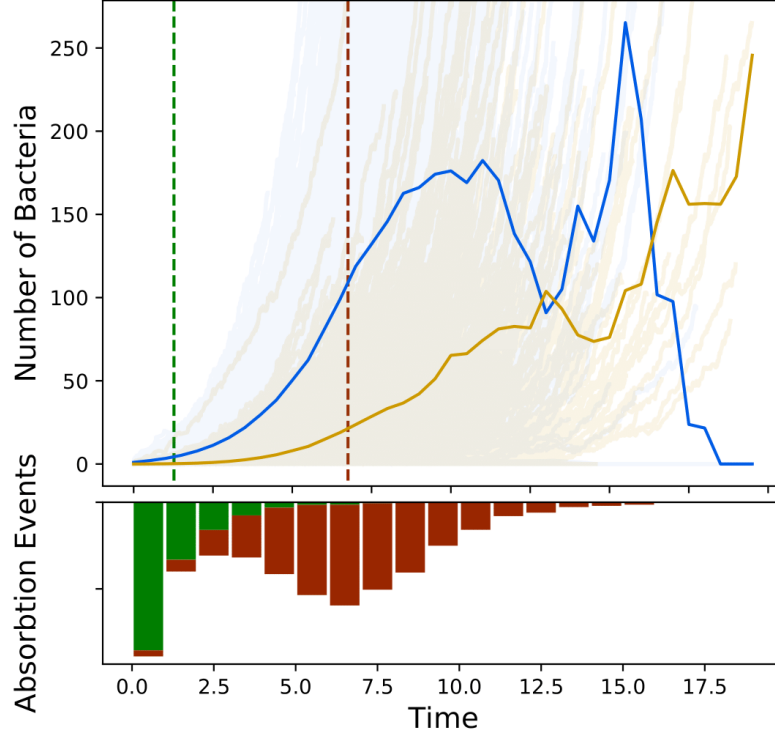


Figure 6: Here we have the results of 5000 iterations of the Python simulation. This experiment differs from the previous one shown in figures 4 & 5 not only in the number of episodes, but also in the fact that, here, the total number of bacteria was not capped, instead being allowed to free roam to higher values. (In practicality, the cap was simply increased to the unreachable ceiling of 50000). The main purpose of presenting this is to offer a clear explanation for the volatile fluctuations that are found in the average number of type b , $\bar{n}_b$, seen at the later times in figure 4: Helped by the fact that all 5000 of the individual simulations completed are projected, visible as the upward reaching wispy lines in the background, we can observe that during the later times there are significantly fewer episodes that have not already terminated. This has the troublesome effect (especially at Time > 15) of limiting the available pool of data by which an average value can be calculated – thus reducing the accuracy of the estimate, and causing the visible fluctuations.

5 Markov chain approach to modelling

As in the section above on numerical approximation using Python, we here consider dynamics of the model in relation to a single macrophage.

Changes in a macrophage, including the numbers of type a & b *Y. pestis* can be described by a continuous time Markov process. Excluding the state in which the macrophage has ruptured, we would have a bi-variate

Markov process,

$$\mathcal{X} = \{\mathbf{X}(t) = (X(t), Y(t)) : t \geq 0\},$$

where $(X(t), Y(t)) = (n_a, n_b)$. Including the rupture state, which will be labelled R , we find that the dynamics are resident to the following state space.

$$\mathcal{S} = \{(n_a, n_b) | n_a \in \{0, \dots, N\}, n_b \in \{0, \dots, N\} \text{ and } n_a + n_b \leq N\} \cup \{R\}.$$

This has two absorbing states: rupture, R , and macrophage recovery, $(n_a, n_b) = (0, 0)$.

Transitions between these states are confined to those laid out in section 3, and will occur with probabilities dependant on the associated rates. For example, the rate with which $(n_a, n_b) = (2, 0)$ will transition into $(3, 0)$ is $\lambda_1^{(2)} = 2\lambda_a$ (2 in line with the linear relationship between birth rate and number of candidate parents). That is to say, the probability of this transition (call it $p_{(0,2),(0,3)}$) occurring during a time interval Δt is $\lambda_a^{(2)} \Delta t$ – assuming no other events happen. If we do not wish to make this assumption of exclusivity, calculating such a probability would require also considering multiple-state-transition “paths” that begin at $(2,0)$ and end at $(3,0)$. Although, due to a handy fact that the occurrence probability of such a multi-path transition will contain a multiple, Δt to the power of the number of sub-transitions involved (it will be $O(\Delta t^\alpha)$, where α is the number of sub-transitions), we find the following. Even without assuming only a single move, the above transition probability will in fact be $\lambda_a^{(2)} \Delta t$ by leading order, provided $\Delta t \rightarrow 0$; otherwise stated, $p_{(0,2),(0,3)} = \lambda_a^{(2)} \Delta t + o(\Delta t)$

Figure 7 offers a visualisation of the state space, $\mathcal{S}$, and the possible transitions which are coloured according to their type.

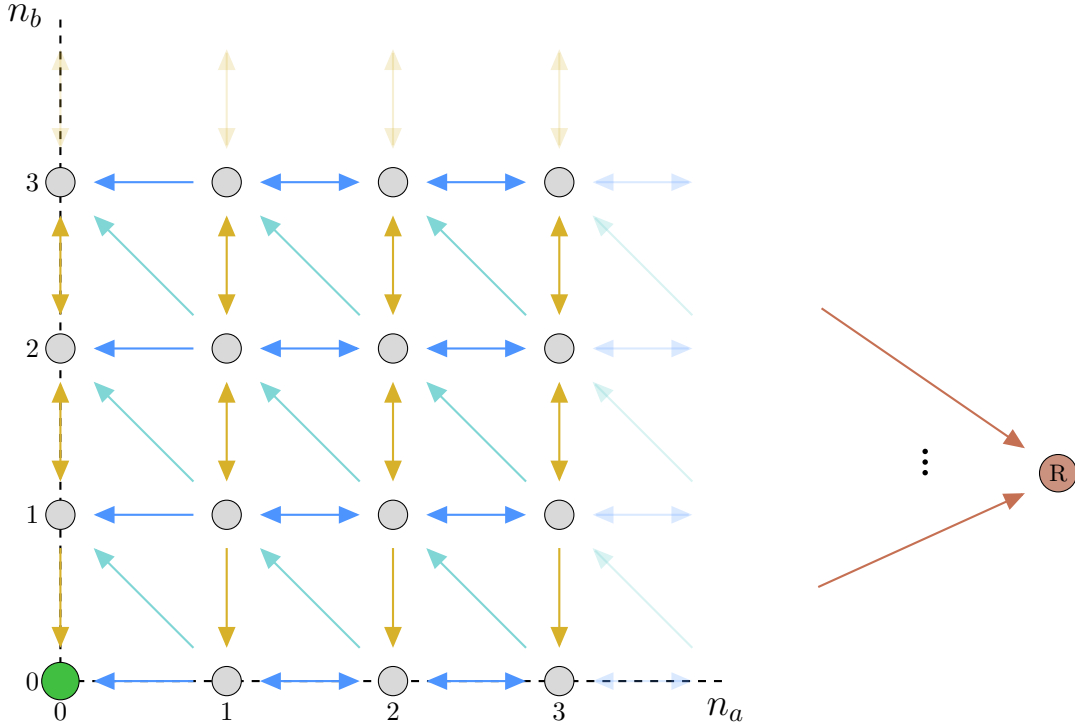


Figure 7: State space $\mathcal{S}$ with Ruptured state R in red on the right; green recovery state is seen at $(0,0)$. Arrows show possible transitions between states: blue for births and deaths of bacteria type a , those of type b are represented by orange, cyan for differentiations from type a to b , and red for the macrophage rupture event.

Moving on, we can define the process by its 1 step transition probabilities,

$$p_{(n_a, n_b), x}(\Delta t) = \mathcal{P}(\mathbf{X}(t + \Delta t) = x | \mathbf{X}(t) = (n_a, n_b)),$$

for states $(n_a, n_b), x \in \mathcal{S}$. Considering each of the various steps that can be taken from an arbitrary state (n_a, n_b) these are as follows.

$$p_{(n_a, n_b), x}(\Delta t) = \begin{cases} n_a \lambda_a \Delta t + o(\Delta t) & x = (n_a + 1, n_b) \\ n_a \mu_a \Delta t + o(\Delta t) & x = (n_a - 1, n_b) \\ n_b \lambda_b \Delta t + o(\Delta t) & x = (n_a, n_b + 1) \\ n_b \mu_b \Delta t + o(\Delta t) & x = (n_a, n_b - 1) \\ n_a \gamma \Delta t + o(\Delta t) & x = (n_a - 1, n_b + 1) \\ n \delta \Delta t + o(\Delta t) & x = R \\ 1 - ((\lambda_a + \mu_a + \gamma + \delta)n_a + (\lambda_b + \mu_b + \delta)n_b)\Delta t + o(\Delta t) & x = (n_a, n_b) \end{cases}$$

(for $\Delta t \rightarrow 0$).

5.1 Late Time Behaviour

If a Markov process has a finite number of states, and from each position it is possible to access every other state by some direct or lengthy path, then if we allow the dynamics to run for infinite time, the process is certain to wander over the full set of positions, however large. Although, this property of freedom is not uniformly present. Within the total set of states, there may well be one or more absorbing subsets. These are defined such that once the process enters, it can never leave. And given their expansively wandering nature mentioned above, provided the existence of one of these trapping zones, a Markov process is sure to become confined at some point prior to infinite time.

It may be seen in some cases that an absorbing subset is resident to a smaller internal absorbing subset, in which case a temporally unrestrained wander, once caught in the “outer” area of confinement is also certain to fall prey to restraint of the “inner”. But this point is superfluous to our present analysis. It may also be seen that an absorbing subset contains only one element – in this case it is called an absorbing state.

Our process indeed has a finite number of states,

$$\frac{N}{2}(N + 1) + 1,$$

and two absorbing states – rupture and recovery – which are freely accessible from the full set of remaining states.

Since we wish to understand the success likelihood of a plague infection in the body, it seems pertinent to consider which of these absorbing states the process is likely to end up in given its initial state. Specifically, given initial state (n_a, n_b) , what is the probability that the macrophage will rupture before it recovers,

$$p_{n_a, n_b}^{(R)};$$

and what is the probability that it will recover, escaping rupture

$$p_{n_a, n_b}^{(0)}?$$

It will be seen that the method for calculating each of these descriptors is similar in nature. So for now let us exclusively examine the likelihood of eventual recovery.

5.1.1 Recovery Probability

The tool of choice for getting at $p_{n_a, n_b}^{(0)}$ will be first step analysis. This involves breaking down the recovery probability at (n_a, n_b) into that after each of the various possible first steps of the system have happened weighted by their proportional likelihood of happening:

$$p_{n_a, n_b}^{(0)} = \frac{\lambda_a^{(n_a)}}{\phi} p_{n_a+1, n_b}^{(0)} + \frac{\mu_a^{(n_a)}}{\phi} p_{n_a-1, n_b}^{(0)} + \frac{\lambda_b^{(n_b)}}{\phi} p_{n_a, n_b+1}^{(0)} + \frac{\mu_b^{(n_b)}}{\phi} p_{n_a, n_b-1}^{(0)} + \frac{\gamma^{(n_a)}}{\phi} p_{n_a-1, n_b+1}^{(0)}$$

where, $\phi_{n_a, n_b} = \lambda_a^{(n_a)} + \mu_a^{(n_a)} + \lambda_b^{(n_b)} + \mu_b^{(n_b)} + \gamma^{(n_a)} + \delta^{(n)}$. Notice there is no term above relating to an initial step of macrophage rupture. This is due to the fact that the probability of recovery after the macrophage has ruptured is necessarily 0.

Then, using the above form we can consider a linear system of equations, each for a $p_{n_a, n_b}^{(0)}$ with a different initial state (n_a, n_b) to obtain the (recovery probability) first step analysis matrix equation, $\mathbf{p}^{(0)} = \mathbf{A} \cdot \mathbf{p}^{(0)} + \mathbf{c}$;

$$\begin{bmatrix} p_{0,1}^{(0)} \\ p_{1,0}^{(0)} \\ p_{0,2}^{(0)} \\ p_{1,1}^{(0)} \\ p_{2,0}^{(0)} \\ p_{0,3}^{(0)} \\ p_{1,2}^{(0)} \\ p_{2,1}^{(0)} \\ p_{3,0}^{(0)} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{\lambda_b^{(1)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & \frac{\lambda_a^{(1)}}{\phi} & 0 & 0 & 0 & 0 \\ \frac{\mu_b^{(2)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_b^{(2)}}{\phi} & 0 & 0 & 0 \\ \frac{\mu_a^{(1)}}{\phi} & \frac{\mu_b^{(1)}}{\phi} & \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & \frac{\lambda_b^{(1)}}{\phi} & \frac{\lambda_a^{(1)}}{\phi} & 0 \\ 0 & \frac{\mu_a^{(2)}}{\phi} & 0 & \frac{\gamma^{(2)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_a^{(2)}}{\phi} \\ 0 & 0 & \frac{\mu_b^{(3)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\mu_a^{(1)}}{\phi} & \frac{\mu_b^{(2)}}{\phi} & 0 & \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\mu_a^{(2)}}{\phi} & \frac{\mu_b^{(1)}}{\phi} & 0 & \frac{\gamma^{(2)}}{\phi} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\mu_a^{(3)}}{\phi} & 0 & 0 & \frac{\gamma^{(3)}}{\phi} & 0 \end{bmatrix} \begin{bmatrix} p_{0,1}^{(0)} \\ p_{1,0}^{(0)} \\ p_{0,2}^{(0)} \\ p_{1,1}^{(0)} \\ p_{2,0}^{(0)} \\ p_{0,3}^{(0)} \\ p_{1,2}^{(0)} \\ p_{2,1}^{(0)} \\ p_{3,0}^{(0)} \end{bmatrix} + \begin{bmatrix} \frac{\mu_b^{(1)}}{\phi} \\ \frac{\mu_a^{(1)}}{\phi} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The (i, j) th element of $\mathbf{A}$ pertains to a transition from state i to j , where the ordering of the states tied to i & j follow a particular pattern. One that moves from point to point, south-eastwards along diagonal lines of successively increasing n . It starts on the $n = 1$ linear state-set, first with point $(n_a, n_b) = (0, 1)$, goes down and right to $(1, 0)$, then, proceeds jumping to the top-left point of the group with $N = 2 - (0, 2)$ – continuing so on and so forth in this pattern.

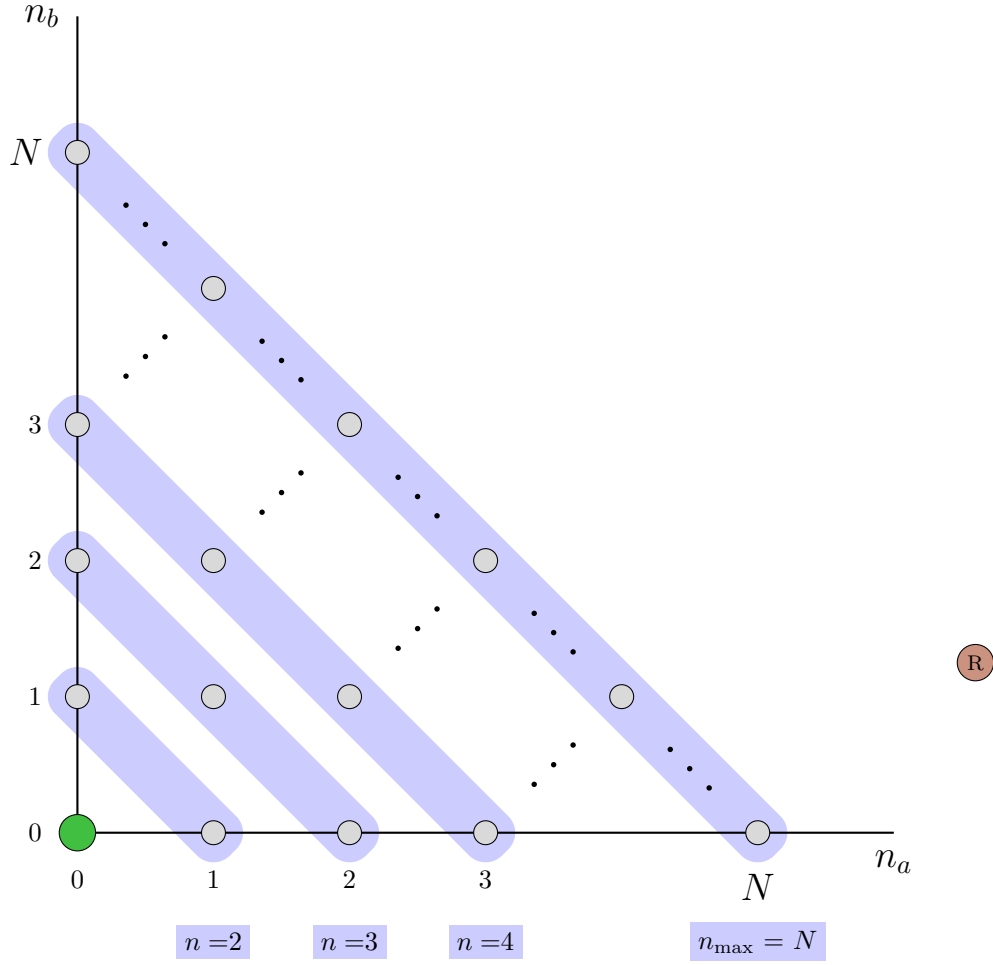


Figure 8: Displays the lines of equal $n_a + n_b = n$ that are successively traversed by the ordering implemented.

Keeping with this idea of “levels” – groups of (n_a, n_b) states that share a particular total number of bacteria, n – let us begin to consider the structure of this matrix $\mathbf{A}$. Take this ordering of the state space we have chosen in constructing $\mathbf{A}$, and the first step matrix equation. Let us divide it by our “levels” as follows.

$$(0, 1) (1, 0) \left| (0, 2) (1, 1) (2, 0) \right| (0, 3) (1, 2) (2, 1) (3, 0) \left| \dots, \right.$$

enforced for each index of the matrix:

$$\mathbf{A} = \left(\begin{array}{cc|cc|cc|cc|cc|cc|cc|cc} 0 & 0 & \frac{\lambda_b^{(1)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & \frac{\lambda_a^{(1)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \frac{\mu_b^{(2)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_b^{(2)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\mu_a^{(1)}}{\phi} & \frac{\mu_b^{(1)}}{\phi} & \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & \frac{\lambda_b^{(1)}}{\phi} & \frac{\lambda_a^{(1)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\mu_a^{(2)}}{\phi} & 0 & \frac{\gamma^{(2)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_a^{(2)}}{\phi} & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & \frac{\mu_b^{(3)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\lambda_b^{(3)}}{\phi} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\mu_a^{(1)}}{\phi} & \frac{\mu_b^{(2)}}{\phi} & 0 & \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_b^{(2)}}{\phi} & \frac{\lambda_a^{(1)}}{\phi} & 0 & 0 \\ 0 & 0 & 0 & \frac{\mu_a^{(2)}}{\phi} & \frac{\mu_b^{(1)}}{\phi} & 0 & \frac{\gamma^{(2)}}{\phi} & 0 & 0 & 0 & 0 & \frac{\lambda_b^{(1)}}{\phi} & \frac{\lambda_a^{(2)}}{\phi} & 0 \\ 0 & 0 & 0 & 0 & \frac{\mu_a^{(3)}}{\phi} & 0 & 0 & \frac{\gamma^{(3)}}{\phi} & 0 & 0 & 0 & 0 & 0 & \frac{\lambda_a^{(3)}}{\phi} \\ \hline 0 & 0 & 0 & 0 & 0 & \frac{\mu_b^{(4)}}{\phi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\mu_a^{(1)}}{\phi} & \frac{\mu_b^{(3)}}{\phi} & 0 & 0 & \frac{\gamma^{(1)}}{\phi} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\mu_a^{(2)}}{\phi} & \frac{\mu_b^{(2)}}{\phi} & 0 & 0 & \frac{\gamma^{(2)}}{\phi} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\mu_a^{(3)}}{\phi} & \frac{\mu_b^{(1)}}{\phi} & 0 & 0 & \frac{\gamma^{(3)}}{\phi} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\mu_a^{(4)}}{\phi} & 0 & 0 & 0 & \frac{\gamma^{(4)}}{\phi} & 0 \end{array} \right)$$

Inspecting the above form, we can see that the divided sections appear to exhibit some repeating structures. Firstly, it is notable that if we consider $\mathbf{A}$ to be made up of sub-matrices defined as bounded by the lines above, there is a tridiagonal structure:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} & \mathbf{A}_{2,3} & \ddots & \vdots \\ \mathbf{0} & \mathbf{A}_{3,2} & \mathbf{A}_{3,3} & \ddots & \mathbf{0} \\ \vdots & \ddots & \ddots & \ddots & \mathbf{A}_{N-1,N} \\ \mathbf{0} & \cdots & \mathbf{0} & \mathbf{A}_{N,N-1} & \mathbf{A}_{N,N} \end{bmatrix}.$$

It is also seen that the non-zero sub-matrices, $\mathbf{A}_{i,j}$, share a pattern with others on their respective diagonal. Immediately one may point out that the lower-diagonal sub-matrices, $\mathbf{A}_{k,k-1}$, only contain elements relating to cell deaths; the upper-diagonals, $\mathbf{A}_{k,k+1}$, only to cell births; and the centre diagonals, $\mathbf{A}_{k,k}$, only to $a \rightarrow b$ transfer events. Further inspection reveals that each of these three sub-matrix groups conforms to a particular pattern as one looks down their diagonal. These structures are thus stated generally:

Lower Diagonal

$$\mathbf{A}_{k,k-1} = \begin{bmatrix} \mu_b^{(k)} & 0 & \cdots & 0 \\ \mu_a^{(1)} & \mu_b^{(k-1)} & \ddots & \vdots \\ 0 & \mu_a^{(2)} & \ddots & 0 \\ \vdots & \ddots & \ddots & \mu_b^{(1)} \\ 0 & \cdots & 0 & \mu_a^{(k)} \end{bmatrix} \times \frac{1}{\phi}$$

Upper Diagonal

$$\mathbf{A}_{k,k+1} = \begin{bmatrix} \lambda_b^{(k)} & 0 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_b^{(k-1)} & \lambda_a^{(1)} & 0 & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \lambda_b^{(1)} & \lambda_a^{(k-1)} & 0 \\ 0 & \cdots & 0 & 0 & 0 & \lambda_a^{(k)} \end{bmatrix} \times \frac{1}{\phi}$$

Centre Diagonal

$$\mathbf{A}_{k,k} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ \gamma^{(1)} & 0 & \cdots & 0 & 0 \\ 0 & \gamma^{(2)} & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 \\ 0 & \cdots & 0 & \gamma^{(k)} & 0 \end{bmatrix} \times \frac{1}{\phi}$$

Closer inspection of these structures brings forth an interesting connection: each sub-matrix, $\mathbf{A}_{i,j}$ contains only elements that relate to state changes coming from “level” $n = i$, and going to “level” $n = j$ (see figure 8 to visualise). (Note: although the centre diagonal is defined over all $k \in \{1, \dots, N\}$, the lower diagonal is truncated by $k \geq 2$, and the upper by $k \leq N - 1$).

5.2 Solving the analytic mathematical model

Leading on from the above workings, we now wish to solve the first step analysis matrix equation for $p_{(n_a, n_b)}^{(0)}$ values by exploiting the tridiagonal sub-matrix structure which is present.

Carrying the same splitting-up regime as used in $\mathbf{A}$, in a one dimensional sense, over to the vectors $\mathbf{p}^{(0)}$ and $\mathbf{c}$, the full portioned first step equation is obtained:

$$\begin{bmatrix} \mathbf{p}_1^{(0)} \\ \mathbf{p}_2^{(0)} \\ \mathbf{p}_3^{(0)} \\ \vdots \\ \mathbf{p}_N^{(0)} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} & \mathbf{A}_{2,3} & \ddots & \vdots \\ \mathbf{0} & \mathbf{A}_{3,2} & \mathbf{A}_{3,3} & \ddots & \mathbf{0} \\ \vdots & \ddots & \ddots & \ddots & \mathbf{A}_{N-1,N} \\ \mathbf{0} & \cdots & \mathbf{0} & \mathbf{A}_{N,N-1} & \mathbf{A}_{N,N} \end{bmatrix} \begin{bmatrix} \mathbf{p}_1^{(0)} \\ \mathbf{p}_2^{(0)} \\ \mathbf{p}_3^{(0)} \\ \vdots \\ \mathbf{p}_N^{(0)} \end{bmatrix} + \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{c}_3 \\ \vdots \\ \mathbf{c}_N \end{bmatrix} \quad (1)$$

And just so we're on the same page – for example:

$$\mathbf{p}_2^{(0)} = \left[p_{(0,2)}^{(0)}, p_{(1,1)}^{(0)}, p_{(2,0)}^{(0)} \right]^\top.$$

Utilising this more orderly arrangement, we can bring in a mechanically iterative analytic process to obtain the values of $\mathbf{p}^{(0)}$ for given input parameters.

Breaking down equation (1) into its constituent rows, we find the recursive formula

$$\mathbf{p}_k^{(0)} = \mathbf{A}_{k,k-1} \mathbf{p}_{k-1}^{(0)} + \mathbf{A}_{k,k} \mathbf{p}_k^{(0)} + \mathbf{A}_{k,k+1} \mathbf{p}_{k+1}^{(0)} + \mathbf{c}_k \quad (2)$$

for each row, k , where $2 \leq k \leq N - 1$. In the special case of $k = 1$, the form is the same apart from that the $\mathbf{p}_{k-1}^{(0)}$ term is missing; likewise when $k = N$, the $\mathbf{p}_{k+1}^{(0)}$ element is not present. Equation (2) can also be

represented as,

$$\mathbf{A}_{k,k-1}\mathbf{p}_{k-1}^{(0)} + (\mathbf{A}_{k,k} - \mathbb{1})\mathbf{p}_k^{(0)} + \mathbf{A}_{k,k+1}\mathbf{p}_{k+1}^{(0)} + \mathbf{c}_k = \mathbf{0}.$$

We will refer to this as $f_k(\mathbf{p}_{k-1}^{(0)}, \mathbf{p}_k^{(0)}, \mathbf{p}_{k+1}^{(0)}) = \mathbf{0}$. Also, to further streamline the notation, the following stand-in will be made temporarily.

$$\mathbf{p}_k^{(0)} \rightarrow q_k$$

As sated previously, the dependencies change with k :

$$\begin{aligned} f_1(q_1, q_2) &= 0, & k &= 1 \\ f_k(q_{k-1}, q_k, q_{k+1}) &= 0, & \text{for } k &= 2, \dots, N-1 \\ f_N(q_{N-1}, q_N) &= 0, & k &= N. \end{aligned}$$

Now, the process for computing $\mathbf{p}_k^{(0)} \forall k$ runs as follows.

1. We start by rearranging $f_1(q_1, q_2) = 0$ into $q_1 = q_1(q_2)$.
2. Take $f_2(q_1, q_2, q_3)$ and substitute $f_2(q_1(q_2), q_2, q_3) = 0$, then rearrange for $q_2 = q_2(q_3)$.
3. Repeat this process taking $q_{k-1} = q_{k-1}(q_k)$ from the previous step, substituting it into $f_k(q_{k-1}(q_k), q_k, q_{k+1})$, subsequently finding $q_k = q_k(q_{k+1})$ for use in the next step.
4. Eventually, we reach $f_N(q_{N-1}(q_N), q_N) = f(q_N) = 0$. This equation can be solved for q_N .
5. With q_N , obtain $q_{N-1}(q_N)$, then with that $q_{N-2}(q_{N-1})$, so on and so forth using the previously encountered values of $q_k(q_{k+1})$ all the way back to $q_1(q_2)$. At which point, the problem is solved - $\mathbf{p}_k^{(0)} \forall k$ are found.

Once we have $\mathbf{p}_k^{(0)} \forall k$, each of the vectors can be dissolved into their constituent $p_{n_a, n_b}^{(0)}$ values representing the probability of recovery when starting with n_a type a *Y. pestis*, and n_b type b .

To apply this algorithmic analytical procedure by hand would be very tedious indeed for any even moderately large maximum bacteria number N . So, I decided it would be best to write a Python code that can do it for us. The only sacrifice in this is that the recovery probabilities, $p_{n_a, n_b}^{(0)}$, are given numerically, rather than as an expression in terms of the input parameters.

Implementing this algorithmic process in Python was indeed successful and you can see the results of this in the heat-map plot below. Validity of the outputs was verified using normalised counts from many Gillespie algorithm simulations.

Actually, after devising and implementing this method, I realised it was possible to simply use a standard linear matrix equation solving function built into numpy. This was a little disappointing, but I am somewhat comforted by the fact that the more tedious approach does yield solutions faster, especially when the maximum number of bacteria, N , is increased.

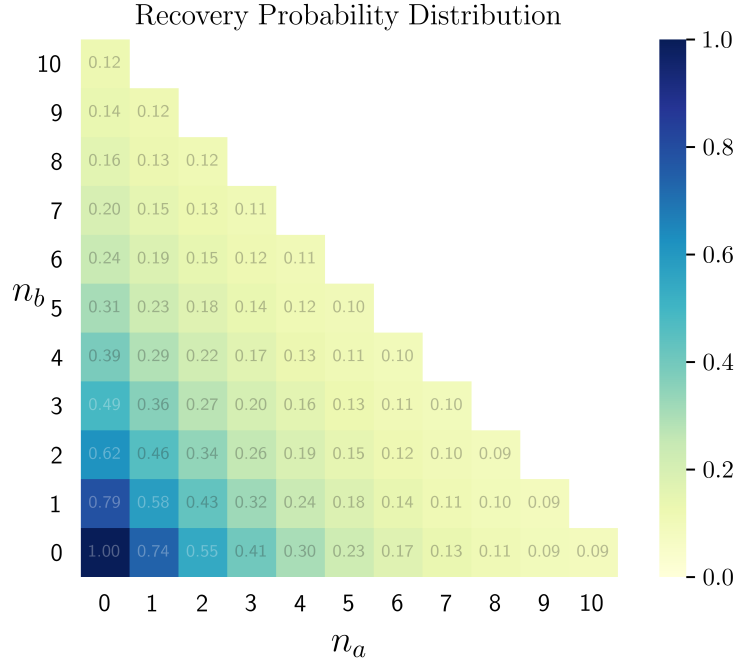


Figure 9: Heat map of recovery probabilities $p_{n_a, n_b}^{(0)}$ for $N = 10$ and the following parameters: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

You will notice that there is a skew in the probabilities with higher values tending to be present in the vicinity of the n_b axis. This can readily be explained by the fact that the birth rate of type a *Y. pestis* bacteria is marginally higher than that of type b . If more of the initial bacteria is of type b , the slower dividing version, likelihood of ultimate survival will be higher.

6 Note on parameters used

Here I will give some rationale for the choice of parameters used in the experiment mentioned above and those which will be discussed subsequently.

An experiment conducted by Spinner et al. [8] showed that contrary to the prevailing notion, *Y. pestis* can survive and divide in neutrophils as well as macrophages. This is cause for consideration in and of itself and will need to be addressed later in the research, but for now, let us discuss their findings insofar as they are relevant to our choice of rate parameter values.

The study found that after 45 minutes a population of within-neutrophil resident *Y. pestis* bacteria had reduced by more than half. At the 22h mark however, this reduced count had instantiated itself to a point greater than that of the initial population size. It seems reasonable to assume that the rate of birth and death of *Y. pestis* in macrophages will be similar to as was seen in these neutrophils. But if we are to use this observation to prescribe a ballpark estimate choice of parameter values to our model, it will first be necessary to draw attention to an important difference between the dynamics observed in this experiment and those that we describe in our mathematical conception: In the experiment, most of the *Y. pestis* killing that took place in the neutrophils appears to have done so early on in the course of observation. This can be interpreted as follows. Some of the initial population of *Y. pestis* were resistant to the neutrophil's intracellular cytotoxic environment; about half were not. Our model paints the dynamical flow of events in a notably distinct way. Birth and death of cells both occur throughout the course of experiment, led by the frequency prescribed to them by exponentially distributed random variables in time. Yes, the rate at which these events occur does change with time along with the increase or reduction in population size, but adjusting for this, the dynamics are temporally uniform – there is no initial purge of sensitive cells.

Taking this difference into consideration makes the choice of parameters and their units necessarily only a rough estimate, but as an initial step, let us propose that a birth rate of $\lambda_a = 1$ be taken in the units of “per half day”. At this, a division event will take place approximately every 12 hours.

For now, we will make the assumption that the birth rate for type b is lower than that of type a – a cost to counter its benefit of resistance to phagocytosis – and that macrophage rupture events occur at a rate about an order of magnitude slower than birth/deaths. The current chosen parameters are as follows: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$.

7 Survival probability functions - an ODE approach

So far we have used a continuous time Markov process model to explore the dynamics of *Yersinia pestis* within a macrophage. We have found, for the full set of possible initial conditions, the probability that a macrophage will ultimately recover (or rupture). In the following section, we will use a different framework to understand dynamics of this system more continually as it progresses. Specifically, we seek the probability that macrophage survival persists beyond some time taken as a continuous variable.

Let us define survival probability functions, $S(t)$ & $G(t)$.

$S(t) = \mathcal{P}[\text{a macrophage is still alive at time } t \mid \text{at time } 0, \text{ it contained a single type } a \text{ Y. pestis bacteria}],$

and $G(t)$ is specified likewise but for an initial state of only a single type b bacteria. Before continuing it is important to mention that this model differs from the one analysed in §5 in an important way: there is no cap N on the maximum possible number of bacteria.

Focusing in on $S(t)$ for the time being, let us consider how its value differs between two instances in time separated by a small interval, Δt . Said plainly, how is $S(t + \Delta t)$ different from $S(t)$? If we take $\Delta t \rightarrow 0$, figuring this out is made simpler by our ability to assume the following. *In the time between t and $t + \Delta t$, if any of the possible events occur, only one will.* Given this, we can consider all of the alternative eventualities independently, and how exactly the value of $S(t + \Delta t)$ is distinct from prior to the realisation of each outcome.

For example, if in the interval of duration Δt , the single permitted event is a cell division, $S(t + \Delta t)$'s value will now be equivalent to $S(t)^2$. In this case, when $t = 0$, $G(0 + \Delta t)$ no longer corresponds with an initial state of a single type a bacteria; instead there are two. Considering a couple of properties of our conception, namely,

- the bacteria operate independently to each other in their host macrophage, and
- the rupture rate is scaled linearly with the number of bacteria,

we can think of this situation as being identical to one in which there are two macrophages, each with a single initial *Y. pestis* bacteria.

The survival likelihood of a single macrophage with two initial bacteria is the same as the probability that two macrophages, each with at first one bacterium will both survive to time t . And with the general multiplication rule of probabilities, this is quantified as $S(t)^2$.

Similar reasoning when applied to the other potential initial steps yields the following updates in survival likelihood:

$$S(t + \Delta t) = \begin{cases} S(t)^2 & \text{when the event is a division of type } a \text{ bacteria,} \\ 1 & \text{when the initial bacteria dies,} \\ G(t) & \text{when it differentiates into type } b, \text{ and} \\ S(t) & \text{for when no event happens.} \end{cases}$$

Each of these outcomes will happen with a probability (listed below) scaled by the relevant rate:

$S(t + \Delta t)$	Associated probability
$S(t)^2$	$\lambda_a \Delta t$
1	$\mu_a \Delta t$
$G(t)$	$\gamma \Delta t$
$S(t)$	$1 - (\lambda_a + \mu_a + \gamma + \delta) \Delta t$

Now, using the contents of the above table we can construct a weighted sum relating $S(t + \Delta t)$ to its alternative equivalent values:

$$S(t + \Delta t) = \lambda_a \Delta t S(t)^2 + \mu_a \Delta t + \gamma \Delta t G(t) + [1 - (\lambda_a + \mu_a + \gamma + \delta) \Delta t] S(t).$$

Then, with some rearrangement, this yields,

$$\frac{S(t + \Delta t) - S(t)}{\Delta t} = \lambda_a S^2 - (\lambda_a + \mu_a + \gamma + \delta) S + \mu_a + \gamma G(t).$$

Given Δt is taken at a limit tending to 0, the LHS of this equation satisfies the definition of a derivative. Thus, we have an ODE for $S(t)$; similar analysis also yields another for $G(t)$, leaving us with the coupled differential equations:

$$\frac{dS}{dt} = \lambda_a S^2 - (\lambda_a + \mu_a + \gamma + \delta) S + \mu_a + \gamma G(t) \quad (3)$$

$$\frac{dG}{dt} = \lambda_b G^2 - (\lambda_b + \mu_b + \delta) G + \mu_b.$$

$G(t)$ is defined analytically as follows (in line with the solution of Jonathan's one-type process in [9]).

$$G(t) = \frac{G_+(G_- - 1) + G_-(1 - G_+)e^{\lambda_b t(G_+ - G_-)}}{(G_- - 1) + (1 - G_+)e^{\lambda_b t(G_+ - G_-)}}$$

Where G_+ & G_- are fixed such that

$$G_+ + G_- = \frac{\lambda_b + \mu_b + \delta}{\lambda_b} \quad G_+ G_- = \frac{\mu_b}{\lambda_b}.$$

As for $S(t)$, the remainder of this section will be set around efforts at getting closer to an analytical description of its form.

Solving the differential equations for S and G numerically provides us with the curves seen in Figure 10. As you can see, both of the survival probabilities monotonically decrease with time; with each, the decline is at first quite steep, but steadily they settle to convergence at constant values. For the parameters used, these late time values, S_- and G_- respectively are 0.738... and 0.787.... And these values line up with those obtained by the Markov chains analysis in §5.

7.1 Hypergeometric function method approach

Similar to as is done by Antal and Krapivsky in [10], we first tried to obtain $S(t)$ analytically using a method involving hypergeometric functions. For this (3) is transformed into a second order ODE by a following trick that is standardly used on Riccati equations of this type. Then an attempt was made to reorder the result

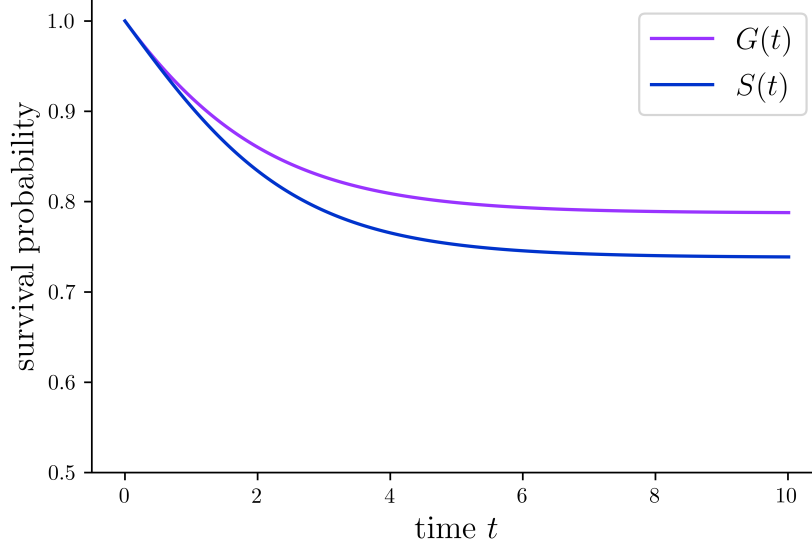


Figure 10: Survival probability functions $S(t)$ and $G(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

of this into the canonical hypergeometric equation:

$$t(1-t)\frac{d^2f}{dt^2} + [c - (a+b+1)t]\frac{df}{dt} - abf = 0.$$

Which, if constructed, automatically confers a pair of solutions which are made up with versions of the hypergeometric series function,

$${}_2F_1(a, b; c; t) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{t^n}{n!}.$$

Unfortunately, the attempt was not successful. Our equation has a more complex inhomogeneous term ($\gamma G(t)$) than the one worked on in [10], and not only this, but their inhomogeneous term was of a structure that allowed for a particular resolving step.

Beginning with (3), for simplicity, we set $\xi = (\lambda_a + \mu_a + \gamma + \delta)$ so that what we have to work with becomes

$$\frac{dS}{dt} = \lambda_a S^2 - \xi S + \mu_a + \gamma G(t). \quad (4)$$

Due to the squared term in S and the fact that it is a first order ODE, this aligns with the specification of a Riccati equation. As pointed to in [10], these are well approached in the following way. Substitute

$$S = \frac{k}{z} \frac{dz}{dt}, \quad (5)$$

where k is an arbitrary constant to be fixed. (4) becomes,

$$\frac{k}{z} \frac{d^2z}{dt^2} - \frac{k}{z^2} \left(\frac{dz}{dt} \right)^2 = \frac{\lambda_a k^2}{z^2} \left(\frac{dz}{dt} \right)^2 - \frac{\xi k}{z} \frac{dz}{dt} + \mu_a + \gamma G(t).$$

By setting $k = -\frac{1}{\lambda_a}$, we can eliminate the nonlinear terms of multiple $\left(\frac{dz}{dt}\right)^2$.

$$-\frac{1}{\lambda_a z} \frac{d^2 z}{dt^2} + \left(\frac{1}{\lambda_a z^2} - \frac{1}{\lambda_a z^2} \right) \left(\frac{dz}{dt} \right)^2 \overset{0}{=} \frac{\xi}{\lambda_a z} \frac{dz}{dt} + \mu_a + \gamma G(t)$$

Leaving, on multiplication by $-\lambda_a z$,

$$\frac{d^2 z}{dt^2} + \xi \frac{dz}{dt} + \lambda_a (\mu_a + \gamma G(t)) z = 0.$$

From here, Antal and Krapivsky considered the limit of $t \rightarrow \infty$. Their inhomogeneous term, taken at this limit, like ours, becomes a constant. This permits the equation to be treated as if it were linear, using $z \propto e^{-\omega t}$ to obtain the characteristic equation,

$$\omega^2 - \xi \omega + \lambda_a (\mu_a + \gamma G_-) = 0.$$

But here arises the bottleneck through which Antal and Krapivsky's simpler inhomogeneous term allowed clear passage, and that forces our analysis off along the wall in search of another way through. They use $z \propto e^{-\omega t}$ in combination with the fact that their term is a basic multiple of " $e^{-\lambda_2 t}$ " to inform the search for a solution of the structure $\hat{z}^{\omega/\lambda_2} \Phi(\hat{z})$ (note: in the paper, [10], $\hat{z}$ appears as z , but I've changed it so as not to overlap with the notation used here). The same move doesn't work with this line of analysis because $G(t)$ is not a basic multiple of e to some power that is a multiple of t . A reasonable exploration was carried out around the area to seek some alternative manipulation, but nothing was found. This is not to say, however, that the avenue is to be completely dropped as possible way on.

7.2 Approximation of $S(t)$

Following limited success in the head-first analytical attempt at finding $S(t)$, we began to explore means of approximation. The first step in this direction was to, fix $G(t)$ in (3) to its late time value G_- ,

$$\frac{dS_0}{dt} = \lambda_a S_0^2 - \xi S + \mu_a + \gamma G_-$$

which simplifies the equation sufficiently for a straightforward solution. Gained from this, $S_0(t)$, conforms with the behaviour of $S(t)$ for time extending toward infinity. It is also fixed such that it begins identically at $S_0(0) = 1$. The analytic formula is as follows and is found via the same reasoning that brings about the form of $G(t)$.

$$S_0(t) = \frac{S_+(1 - S_-) - S_-(1 - S_+)e^{\lambda_a(S_+ - S_-)t}}{(1 - S_-) - (1 - S_+)e^{\lambda_a(S_+ - S_-)t}},$$

where,

$$S_{\pm} = \frac{\zeta}{2} \pm \sqrt{\frac{\zeta^2}{4} - \mu_a + \gamma G_-}, \quad \text{with} \quad \zeta = \frac{\lambda_a + \mu_a + \gamma + \delta}{\lambda_a}.$$

7.3 Disparity measure, $R(t)$

Beyond this, we became interested in the extent to which the disparity between survival likelihood, S , and its approximation, S_0 , persists throughout time. It became convenient to ascribe the letter R to this difference

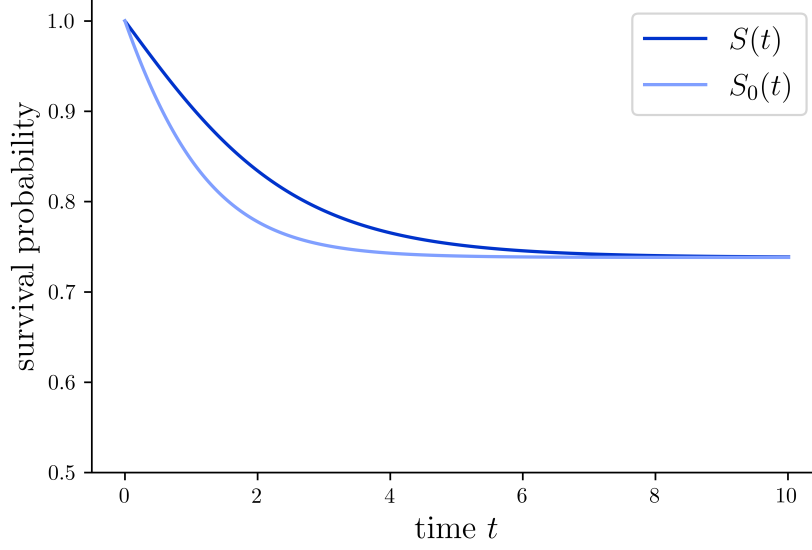


Figure 11: Survival probability function $S(t)$ and its approximation $S_0(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

such that

$$R(t) = S - S_0.$$

From the plot below, we can see that $R(t)$ both begins at and tends toward 0; there is only a portion, during which S and S_0 are still in flux, where the disparity sharply rises up to a maximum value, then steadily recedes.

Given our lack of an analytic description of survival probability $S(t)$, we also do not yet possess a formula for the shortcoming $R(t)$. But what can we say rigorously about its behaviour? The following line of approach tries to constrain the path of $R(t)$ between an upper, and lower bound. There are many possible curves which satisfy the condition of being uniformly above, or indeed below $R(t)$ for all $t > 0$. The question is: how narrow of an interval can we describe precisely that R will conform to?

ODE for $R(t)$

In preparation for this search, let us lay out the equation that describes the variations of R through time. Taking (3) in combination with the relation of near identical form that dictates S_0 , it may be seen that

$$\frac{dR}{dt} = \frac{d}{dt}(S - S_0) = \lambda_a(S^2 - S_0^2) - (\lambda_a + \mu_a + \gamma + \delta)R + \gamma(G - G_-).$$

Which can be rearranged into

$$\frac{dR}{dt} = [\lambda_a(S + S_0 - 1) - (\mu_a + \delta + \gamma)]R + \gamma(G - G_-). \quad (6)$$

7.4 Proof of boundedness

Before getting fully into the analysis associated with finding curves that bound $R(t)$ from above and below, it will be helpful to understand a behaviour captured by the following theorem.

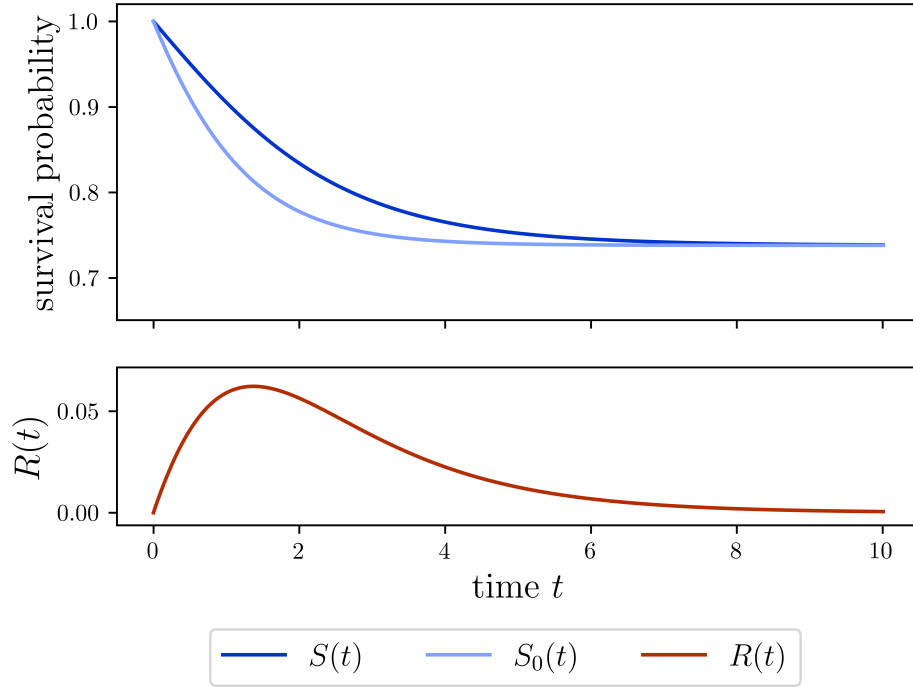


Figure 12: Survival probability function $S(t)$, its approximation $S_0(t)$, and the difference between them, $R(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

Theorem 1. *With the following equations describing the behaviour of $y(t)$ and $\tilde{y}(t)$,*

$$\frac{dy}{dt} = \alpha(t)y + \beta(t), \quad \frac{d\tilde{y}}{dt} = \tilde{\alpha}(t)\tilde{y} + \tilde{\beta}(t). \quad (7, 8)$$

Given $\alpha(t) > \tilde{\alpha}(t)$ & $\beta(t) > \tilde{\beta}(t)$, $\forall t > 0$; $\alpha(0) = \tilde{\alpha}(0)$ & $\beta(0) = \tilde{\beta}(0)$; and, $y(0) = \tilde{y}(0) \geq 0$. For all positive time ($t > 0$),

$$y(t) > \tilde{y}(t).$$

Prior to engaging the proof, it will be well worth pausing to note that this theorem can be looked at and used in “both directions”. It can equally be read to declare $\tilde{y}$ as a lower bound of y for $t > 0$, or indeed, y as an upper bound of $\tilde{y}$.

Proof. $y' = \alpha(t)y + \beta(t)$ can be re-written using the definition of a derivative:

$$y(t + \Delta t) = y(t) + [\alpha(t)y + \beta(t)]\Delta t, \quad \text{as } \Delta t \rightarrow 0$$

Without loss of generality, this can be considered to be the description of a sequence,

$$y_0(0), y_1(\Delta t), y_2(2\Delta t), \dots, y_n(n\Delta t).$$

And provided Δt is taken at a limit approaching 0 from above, this sequence, mapped also for a number of

instances, n , which extends unendingly towards infinity, will dictate the behaviour of y identically as is done by (7). Our sequence relation thus becomes

$$y_{n+1}(t_{n+1}) = y_n(t_n) + [\alpha(t_n)y_n(t_n) + \beta(t_n)]\Delta t,$$

where $t_n = n\Delta t$, $\Delta t \rightarrow 0$ and $n \rightarrow \infty$. Which is equivalent to

$$y_{n+1}(t_{n+1}) = y_n(t_n)[1 + \alpha(t_n)\Delta t] + \beta(t_n)\Delta t. \quad (9)$$

Likewise can be done for (8) to bring about

$$\tilde{y}_{n+1}(t_{n+1}) = \tilde{y}_n(t_n)[1 + \tilde{\alpha}(t_n)\Delta t] + \tilde{\beta}(t_n)\Delta t. \quad (10)$$

We will now demonstrate by induction that for all $n \in \mathbb{Z}^+ \cup \{0\}$,

$$y_n \geq \tilde{y}_n.$$

A strict inequality will emerge thereafter. To start, consider the base case of $n = 0$. It is clear that given $y_0(0) = \tilde{y}_0(0) \geq 0$, indeed

$$\begin{aligned} y_1 &= y_0[1 + \alpha(t_0)\Delta t] + \beta(t_0)\Delta t \geq \tilde{y}_0[1 + \tilde{\alpha}(t_0)\Delta t] + \tilde{\beta}(t_0)\Delta t = \tilde{y}_1 \\ y_1 &\geq \tilde{y}_1; \end{aligned}$$

in fact, $y_1 = \tilde{y}_1$ because $\alpha(0) = \tilde{\alpha}(0)$ & $\beta(0) = \tilde{\beta}(0)$.

Let us also observe the second case ($n = 1$) – the reason for this will soon become clear. Similarly,

$$\begin{aligned} y_2 &= y_1[1 + \alpha(t_1)\Delta t] + \beta(t_1)\Delta t \geq \tilde{y}_1[1 + \tilde{\alpha}(t_1)\Delta t] + \tilde{\beta}(t_1)\Delta t = \tilde{y}_2 \\ y_2 &\geq \tilde{y}_2. \end{aligned} \quad (11)$$

But at this index, because $\beta(t_1) > \tilde{\beta}(t_1)$, we can say more strictly that $y_2 > \tilde{y}_2$.

Now, as an induction hypothesis, assume $y_k > \tilde{y}_k$ for some $k \geq 2$. Along the same line of reasoning associated with (11),

$$\begin{aligned} y_{k+1} &= y_k[1 + \alpha(t_k)\Delta t] + \beta(t_k)\Delta t > \tilde{y}_k[1 + \tilde{\alpha}(t_k)\Delta t] + \tilde{\beta}(t_k)\Delta t = \tilde{y}_{k+1} \\ y_{k+1} &> \tilde{y}_{k+1}. \end{aligned}$$

Hence by induction $y_n \geq \tilde{y}_n \forall n \in \mathbb{Z}^+ \cup \{0\}$. But more specifically, $y_n = \tilde{y}_n$ for $n = 0, 1$, and $y_n > \tilde{y}_n$ for $n \geq 2$.

In the process of reconstituting this discrete sequential conception, (9, 10), back into its continuous-time antecedent, (7, 8), we must impose the limit $\Delta t \rightarrow 0$. And in the context of this limit, we are open to make the following translation: $t_{n+1} \rightarrow t_n$, because as $\Delta t \rightarrow 0$, $t_n = n\Delta t$ and $t_{n+1} = n\Delta t + \Delta t$ drift ever closer towards equivalence.

Then, framed as such, we are able to state that for any time t greater than 0,

$$y(t) > \tilde{y}(t).$$

□

7.5 An upper bound on $R(t)$

We will now derive an upper bound on $R(t)$, calling it $R_U(t)$. Given G , S , and S_0 all describe survival probabilities that are monotonically decreasing from 1 with time, it stands true that

$$S < 1, \quad S_0 < 1 \quad \& \quad G < 1, \quad \forall t > 0.$$

Therefore, if we substitute 1 into the positions of these respective functions in equation (6), because the coefficients of their placements are all positive, such a move will only increase the value of both what multiplies R in the RHS and the constant term which follows. In other words, when

$$\frac{dR}{dt} = \tilde{\mathcal{K}}(S(t), S_0(t))R + \tilde{\mathcal{Q}}(G(t))$$

becomes

$$\frac{dR_U}{dt} = \mathcal{K}(S = 1, S_0 = 1)R_U + \mathcal{Q}(G = 1),$$

because the multiplying constants of G , S , and S_0 are all positive, we have

$$\begin{aligned} \mathcal{K}(S = 1, S_0 = 1) &> \tilde{\mathcal{K}}(S(t), S_0(t)) \quad \& \\ \mathcal{Q}(G = 1) &> \tilde{\mathcal{Q}}(G(t)), \end{aligned}$$

$\forall t > 0$. Which neatly lines up with the conditions of Theorem 1, thus allowing us to state that

$$R_U(t) > R(t), \quad \forall t > 0.$$

In its newly simplified form, the equation,

$$\frac{dR_U}{dt} = -(\mu_a + \delta + \gamma - \lambda_a)R_U + \gamma(1 - G_-)$$

admits the straightforward solution by

$$R_U(t) = \frac{\gamma(1 - G_-)}{k} \left(1 - e^{-kt} \right), \tag{12}$$

where $k = \mu_a + \delta + \gamma - \lambda_a$. And this behaves as follows.

As is seen, $R_U(t)$ will tend to some limit, $\gamma(1 - G_-)/k$, as $t \rightarrow \infty$, but only as long as k is positive. Or more specifically provided $\mu_a + \delta + \gamma > \lambda_a$; the birth rate of type a bacteria cannot be too large in relation to the rates of the three events which hinder such a population – death, differentiation, and environment collapse.

7.6 Deriving a lower bound on $R(t)$, $R_L^{(1)}$

Similar to that above, we can also make substitutions into equation (6) that serve to undercut its terms, subsequently providing a solution curve that bounds $R(t)$ from below. One may think to simply substitute S , S_0 , and G with 0, as given these functions all remain positive with time, such a move would indeed provide a lower bound. But we can do better than that. Ideally, we want to make substitutions of S , S_0 , and G that will remain below their values with time, but will also fit as closely as possible to the original curves. Our only constraint in this aim is that the resulting equation must be analytically tractable.

For this lower bound calculation, we will replace S and S_0 by the value they tend down to at late time, S_- . Initially, we tried keeping $G(t)$ as it was, but due to its functional form, even a reduction in complexity

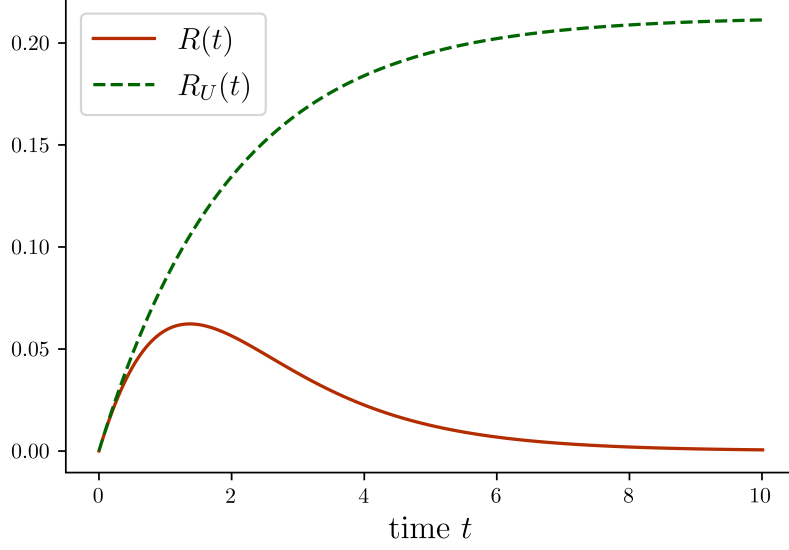


Figure 13: $R(t)$ and its upper bound, $R_U(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

this far doesn't admit the equation with a straightforward solution process. Instead, we swap $G(t)$ out with a different function,

$$f_L^{(1)}(t) = G_- + (1 - G_-)e^{-u_0 t},$$

where $u_0 = \lambda_b(G_+ - G_-)$. $f_L^{(1)}$ is convenient for our purpose on multiple counts: At $t = 0$ and $t \rightarrow \infty$, it conforms neatly with the behaviour of $G(t)$; but importantly, despite sharing a late-time limit, it is true that $f_L^{(1)}(t) < G(t) \forall t > 0$. And this is quite handily proveable (see Appendix A.1). Crucially though, substituting $G(t)$ with $f_L^{(1)}$ does in fact free-up the equation into tractability. Let us now solve it.

Following the aforementioned substitutions, we are left with

$$\frac{dR_L^{(1)}}{dt} = -(\mu_a + \delta + \gamma)R_L^{(1)} + \lambda_a(2S_- - 1)R_L^{(1)} + \gamma(f_L^{(1)} - G_-).$$

Which can also be written as,

$$\frac{dR_L^{(1)}}{dt} = -k_1 R_L^{(1)} + \gamma(1 - G_-)e^{-u_0 t} \quad (13)$$

where $k_1 = \mu_a + \delta + \gamma + \lambda_a(1 - 2S_-)$. Using the normal method of an integration factor, (13) reduces to

$$e^{k_1 t} R_L^{(1)} = \gamma(1 - G_-) \int e^{(k_1 - u_0)t} dt.$$

And with the initial condition $R_L^{(1)}(0) = 0$, this obtains the solution,

$$R_L^{(1)} = \frac{\gamma(1 - G_-)}{k_1 - u_0} \left(e^{-u_0 t} - e^{-k_1 t} \right),$$

which describes the curve seen in Figure 17. It is clear that $R_L^{(1)}(t)$ serves as a lower bound to the difference

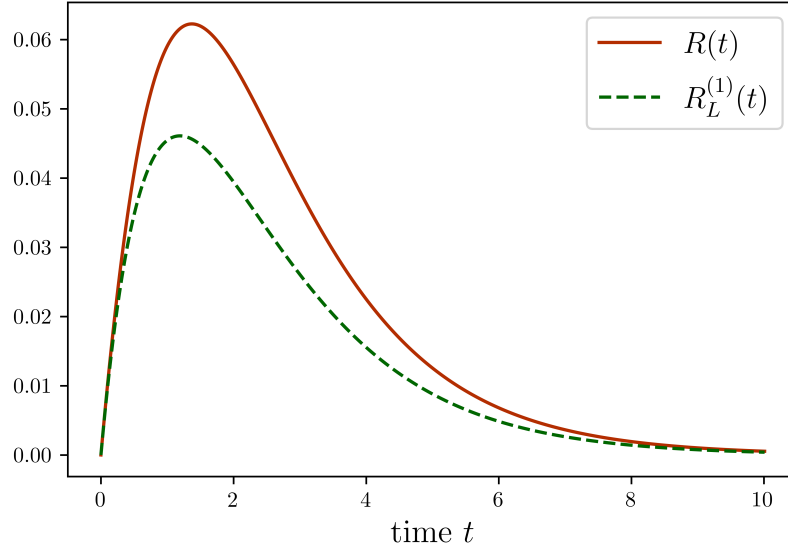


Figure 14: $R(t)$ and its lower bound, $R_L^{(1)}(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

measure, $R(t)$, but can we do better? Will it be possible to find another lower bound with a closer fit to the original? This will be addressed in the next sub-section, but first I will briefly append this one with a result relating to the curve. I will also direct your attention to appendix B where a couple of plots summarise the findings up to this point.

Taking $R_L^{(1)}$'s derivative at 0, it is possible to find the maximum value of this lower bound. The turning point in question comes about at time,

$$t_{\max} = \frac{\log\left(\frac{k_1}{u_0}\right)}{k_1 - u_0},$$

and at disparity value,

$$R_{L \max}^{(1)}(t_{\max}) = \frac{\gamma(1 - G_-)}{k_1 - u_0} \left(\left(\frac{k_1}{u_0} \right)^{\frac{u_0}{u_0 - k_1}} - \left(\frac{k_1}{u_0} \right)^{\frac{k_1}{u_0 - k_1}} \right).$$

With this, along with a result from above, we can say that the peak value of $R(t)$ will occur somewhere in the interval,

$$R_{L \max}^{(1)}(t_{\max}) < R_{\max} < \frac{\gamma(1 - G_-)}{k}.$$

7.7 Deriving a better lower bound, $R_L^{(2)}$

To get a better fitting lower bound on $R(t)$, we will use a convenient fact: $S_0(t)$ is of a very similar form to $G(t)$. Because of this, it proves very simple to generate a function that serves as a lower bound for $S_0(t)$, as

$f_L^{(1)}$ does for $G(t)$. This we call $f_L^{(2)}$, and is described by

$$f_L^{(2)} = S_- + (1 - S_-)e^{-v_0 t},$$

where $v_0 = \lambda_b(S_+ - S_-)$. As with $f_L^{(1)}$ for $G(t)$, $f_L^{(2)}$ both behaves identically to $S_0(t)$ at $t = 0$ & as $t \rightarrow \infty$, and importantly, is always less than $S_0(t)$ for $t > 0$. This is shown by appendix A.2.

Once again, we can use the logic of Theorem 1 by making these substitutions (S_- , $f_L^{(2)}$ and $f_L^{(1)}$ for S , S_0 and G) to create the lower bound for $R(t)$, $R_L^{(2)}$. Following this move, we are faced with the equation,

$$\frac{dR_L^{(3)}}{dt} = -(\mu_a + \delta + \gamma)R_L^{(3)} + \lambda_a \left(S_- + f_L^{(3)} - 1 \right) R_L^{(3)} + \gamma \left(f_L^{(1)} - G_- \right)$$

which, after inclusion of the functional forms and rearrangement, can be written as,

$$\frac{dR_L^{(2)}}{dt} = - \left(k_1 - \lambda_a(1 - S_-)e^{-v_0 t} \right) R_L^{(3)} + \gamma(1 - G_-)e^{-u_0 t},$$

where $k_1 = (\mu_a + \delta + \gamma + \lambda_a(1 - 2S_-))$. Let us now solve for $R_L^{(2)}$.

Taking the $R_L^{(2)}$ multiplied term over to the left, we can apply an integration factor as standard:

$$\frac{d}{dt} \left(e^{H(t)} R_L^{(3)} \right) = \gamma(1 - G_-)e^{H(t)-u_0 t}. \quad (14)$$

The factor being $e^{H(t)}$ such that,

$$H(t) = \int k_1 - \lambda_a(1 - S_-)e^{-v_0 t} dt = k_1 t + \frac{\lambda_a}{v_0}(1 - S_-)e^{-v_0 t}.$$

Continuing from (14) becomes a question of evaluating the following integral.

$$e^{H(t)} R_L^{(3)} = \gamma(1 - G_-) \int e^{H(t)-u_0 t} dt \quad (15)$$

We will now divert on a tangent to this aim, before re-joining the line of reasoning with our findings. For later clarity, let us re-notate our parameters such that $H(t) - u_0 t = a_1 t + a_2 e^{a_3 t}$. The integral in question becomes

$$\int e^{a_1 t + a_2 e^{a_3 t}} dt, \quad (16)$$

where,

$$\begin{aligned} a_1 &= k_1 - u_0, \\ a_2 &= \lambda_a v_0^{-1}(1 - S_-), \text{ \& } \\ a_3 &= -v_0. \end{aligned}$$

At first glance, this integral seems difficult to approach. As it turns out, we can get at it using a couple of series expansions. Specifically, we will be using:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad (x+y)^n = \sum_{m=0}^n \frac{n!}{m!(n-m)!} m^m y^{n-m}. \quad (17, 18)$$

With (17), the integral (16) becomes

$$\int \sum_{n=0}^{\infty} \frac{1}{n!} (a_1 t + a_2 e^{a_3 t})^n dt. \quad (19)$$

Then, applying (18), (19) can be transformed into

$$\int \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} (a_1 t)^m (a_2 e^{a_3 t})^{n-m} dt,$$

equivalent to

$$\int \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} t^m e^{a_3(n-m)t} dt. \quad (20)$$

Moving the integral inside the summations, we are faced with an evaluation that appears more tractable:

$$\sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} \int t^m e^{a_3(n-m)t} dt. \quad (21)$$

Of course, we must notice that the way in which this integral will be resolved depends fundamentally on the particular present values of n and m . Considering the relevant case-domains, we find that

$$\int t^m e^{a_3(n-m)t} dt = \Theta_{n,m}(t) + C$$

where,

$$\Theta_{n,m}(t) = \begin{cases} t & m = n = 0 \\ (m+1)^{-1} t^{m+1} & m = n \neq 0 \\ (a_3 n)^{-1} e^{a_3 n t} & m = 0, n \neq 0 \\ \sum_{l=0}^m \frac{(-1)^l}{(a_3(n-m))^{l+1}} \left(\frac{n!}{(n-l)!} \right) t^{m-l} e^{a_3(n-m)t} & 0 \neq m \neq n \neq 0. \end{cases} \quad (22)$$

The latter of these is reasoned to via a sequential integration by parts approach. At this point we have a solution for (16), only with an integration constant (C) to be fixed:

$$\sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} \Theta_{n,m}(t) + C$$

We can now rejoin the line of reasoning at the place where our tangent departed from, inputting our findings to (23), then dividing by $e^{H(t)}$,

$$R_L^{(2)} = \gamma(1 - G_-)e^{-H(t)} \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} \Theta_{n,m}(t) + C e^{-H(t)}. \quad (23)$$

Using the initial condition, $R_L^{(2)}(0) = 0$, to fix this constant, C , will require some thought on the behaviour of $\Theta_{n,m}$ when $t = 0$. Of its four components, the first two as listed above in (22) simply conform to 0 along with t ; the third part evaluates to a constant, $(a_3(n-m))^{-1}$; and the last group, by far the most interesting, behaves differently at $t = 0$, depending on the particular index, l , in its summation. When $l = 0, \dots, m-1$, or better put, when $m-l \neq 0$, the $t^{m-l}|_{t=0}$ simply nullifies things. But in the final term of the sum when $l = m$, we have a non-zero contribution to $\Theta_{n,m}$ as is seen in the following.

$$\Theta_{n,m}(0) = \begin{cases} 0 & m = n = 0 \\ 0 & m = n \neq 0 \\ (a_3(n-m))^{-1} & m = 0, n \neq 0 \\ \frac{(-1)^m}{(a_3(n-m))^{m+1}} \left(\frac{n!}{(n-m)!} \right) & 0 \neq m \neq n \neq 0. \end{cases} \quad (24)$$

C is prescribed in line with,

$$0 = \gamma(1 - G_-)e^{-H(0)} \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} \Theta_{n,m}(0) + C e^{-H(0)}.$$

Therefore, considering the appropriate (n, m) case domains appropriately with (24), we can isolate C as such

$$C = -\gamma(1 - G_-)e^{-a_2} \left[\sum_{n=1}^{\infty} \frac{1}{n!^2} a_2^n \frac{1}{a_3 n} + \sum_{n=2}^{\infty} \frac{1}{n!} \sum_{m=1}^{n-1} \frac{n!}{m!(n-m)!} a_1^m a_2^{n-m} \frac{(-1)^m}{(a_3(n-m))^{m+1}} \left(\frac{n!}{(n-m)!} \right) \right]$$

Using this, some rearrangement gives the analytic form of $R_L^{(2)}$:

$$R_L^{(2)} = \gamma(1 - G_-)e^{-H(t)} \left\{ \left[\sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m!(n-m)!} a_1^m a_2^{(n-m)} \Theta_{n,m}(t) \right] - \left[\frac{a_2}{a_3} + \sum_{n=2}^{\infty} \left(\frac{1}{n!^2} \cdot \frac{a_2^n}{a_3 n} + n! \sum_{m=1}^{n-1} \frac{a_1^m a_2^{n-m}}{m!(n-m)!} \cdot \frac{(-1)^m}{(a_3(n-m))^{m+1}} \right) \right] \right\}.$$

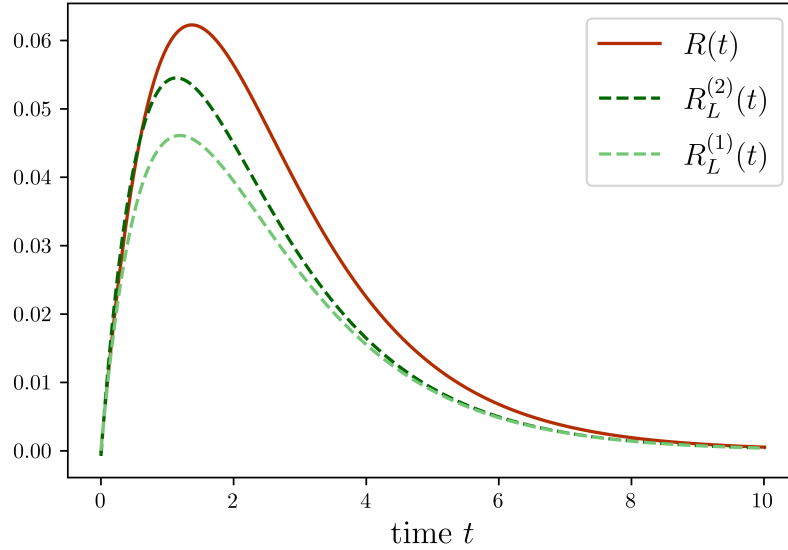


Figure 15: $R(t)$ and its lower bounds, $R_L^{(1)}(t)$ and $R_L^{(2)}(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

As seen in the plot below, $R_L^{(2)}(t)$ fits closer to $R(t)$ than $R_L^{(1)}(t)$. Unfortunately though, in the region of time close to 0, the line dictated by $R_L^{(2)}(t)$ actually rises above the red line of $R(t)$. This is more clear when different parameters are used. I am hoping the culprit for this deviation from what's to be expected is only a typo, but despite some relatively thorough checks, it is yet to be confirmed.

8 Idea for a new model

The following section lays out a style of approach that I plan to explore next. It differs from the previous methods used in a couple of key ways: Instead of focusing in on the behaviour of a *Yersinia pestis* population in a single macrophage, this conception will extend the field of vision to include the dynamics in multiple macrophages – and potentially neutrophils – all resident in some extracellular medium. Additionally to this, rather than working with variables that represent the probability of a chosen outcome, the proposal is to describe the mean quantities of the various types of agent at play.

Those “types of agent” are listed below with their corresponding variable count that is to be thought of as a mean.

$\mathcal{M}$: macrophages

$\mathcal{N}$: neutrophils

E_a : extracellular type a *Yersinia pestis*

E_b : extracellular type b

M_a : within macrophage type a

M_b : within macrophage type b

N_a : within neutrophil type a

N_b : within neutrophil type b

It is worth noting that there is a pooling of intracellular bacteria across the full number of a given type of immune cell; macrophages, for example, aren't accounted for distinctly to harbour an isolated population of bacteria, but are considered collectively. Together they host the class of bacteria that is defined to reside in macrophages.

This particularity in the setup needs to be paid attention when deciding how the variables change at the event of a macrophage rupture. In addition to the number of macrophages decreasing by 1, it appears appropriate to increase the count of extracellular type a *Y. pestis* by exactly the *average* number of type a bacteria in a macrophage. That is, the count of within-macrophage type a divided by the number of macrophages present. The same being done for type b bacteria.

Along with macrophage ruptures, there are many other events that may happen inside immune cells, or floating in the extracellular medium. These are listed below with their respective effects on the variables, prescribed rate parameters, and linearly adjusted effective rates of occurrence.

Event	Result	Rate parameter	Effective rate
Extracellular			
macrophage created	$\mathcal{M} \rightarrow \mathcal{M} + 1$	β_M	β_M
neutrophil created	$\mathcal{N} \rightarrow \mathcal{N} + 1$	β_N	β_N
type a division	$E_a \rightarrow E_a + 1$	λ_a	$\lambda_a E_a$
type b division	$E_b \rightarrow E_b + 1$	λ_b	$\lambda_b E_b$
type a death	$E_a \rightarrow E_a - 1$	μ_a	$\mu_a E_a$
type b death	$E_b \rightarrow E_b - 1$	μ_b	$\mu_b E_b$
Within macrophage environment			
type a absorbtion	$M_a \rightarrow M_a + 1$	α_M	$\alpha_M E_a \mathcal{M}$
type a division	$M_a \rightarrow M_a + 1$	$\tilde{\lambda}_a$	$\tilde{\lambda}_a M_a$
type b division	$M_b \rightarrow M_b + 1$	$\tilde{\lambda}_b$	$\tilde{\lambda}_b M_b$
type a death	$M_a \rightarrow M_a - 1$	$\tilde{\mu}_a$	$\tilde{\mu}_a M_a$
type b death	$M_b \rightarrow M_b - 1$	$\tilde{\mu}_b$	$\tilde{\mu}_b M_b$
macrophage rupture	$E_a \rightarrow E_a + \frac{M_a}{\mathcal{M}}, E_b \rightarrow E_b + \frac{M_b}{\mathcal{M}}, \mathcal{M} \rightarrow \mathcal{M} - 1$ $M_a \rightarrow M_a - \frac{M_a}{\mathcal{M}}, M_b \rightarrow M_b - \frac{M_b}{\mathcal{M}}$	δ_M	$\delta_M (M_a + M_b)$
a differentiates to b	$M_a \rightarrow M_a - 1, M_b \rightarrow M_b + 1$	γ	γM_a
Within neutrophil environment			
type a absorbtion	$N_a \rightarrow N_a + 1$	α_N	$\alpha_N E_a \mathcal{N}$
type a division	$N_a \rightarrow N_a + 1$	$\hat{\lambda}_a$	$\hat{\lambda}_a N_a$
type b division	$N_b \rightarrow N_b + 1$	$\hat{\lambda}_b$	$\hat{\lambda}_b N_b$
type a death	$N_a \rightarrow N_a - 1$	$\hat{\mu}_a$	$\hat{\mu}_a N_a$
type b death	$N_b \rightarrow N_b - 1$	$\hat{\mu}_b$	$\hat{\mu}_b N_b$
neutrophil rupture	$E_a \rightarrow E_a + \frac{N_a}{\mathcal{N}}, E_b \rightarrow E_b + \frac{N_b}{\mathcal{N}}, \mathcal{N} \rightarrow \mathcal{N} - 1$ $N_a \rightarrow N_a - \frac{N_a}{\mathcal{N}}, N_b \rightarrow N_b - \frac{N_b}{\mathcal{N}}$	δ_N	$\delta_N (N_a + N_b)$

This list is only an initial draft of the dynamics to be analysed, the final choice will probably look a bit different. For example, it may be sensible to not work with two types of leukocytes, but only include macrophages. It may also be interesting down the line to include a quantity relating to the level of anti-inflammatory Yop proteins released by *Y. pestis*. This could facilitate an understanding of the biphasal nature of a plague infection, whereby there is an initial phase characterised by no significant immune response, followed by a sudden and intense phase transition to mode of inflammatory action.

Using the effective rates and changes in mean quantity laid out in the table above, it is possible to set up a system of ODEs that describe the behaviour of each variable over time:

$$\begin{aligned}
\frac{d\mathcal{M}}{dt} &= \beta_M - \delta_M(M_a + M_b) \\
\frac{d\mathcal{N}}{dt} &= \beta_N - \delta_N(N_a + N_b) \\
\frac{dE_a}{dt} &= (\lambda_a - \mu_a)E_a + \delta_M(M_a + M_b)\frac{M_a}{\mathcal{M}} + \delta_N(N_a + N_b)\frac{N_a}{\mathcal{N}} \\
\frac{dE_b}{dt} &= (\lambda_b - \mu_b)E_b + \delta_M(M_a + M_b)\frac{M_b}{\mathcal{M}} + \delta_N(N_a + N_b)\frac{N_b}{\mathcal{N}} \\
\frac{dM_a}{dt} &= \alpha_M E_a \mathcal{M} + (\tilde{\lambda}_a - \tilde{\mu}_a)M_a - \delta_M(M_a + M_b)\frac{M_a}{\mathcal{M}} - \gamma M_a \\
\frac{dM_b}{dt} &= (\tilde{\lambda}_b - \tilde{\mu}_b)M_b - \delta_M(M_a + M_b)\frac{M_b}{\mathcal{M}} + \gamma M_a \\
\frac{dN_a}{dt} &= \alpha_N E_a \mathcal{N} + (\hat{\lambda}_a - \hat{\mu}_a)N_a - \delta_N(N_a + N_b)\frac{N_a}{\mathcal{N}} \\
\frac{dN_b}{dt} &= (\hat{\lambda}_b - \hat{\mu}_b)N_b - \delta_N(N_a + N_b)\frac{N_b}{\mathcal{N}}
\end{aligned}$$

It seems reasonable to expect that the behaviour of this system will follow one of two trends. Either the immune cells will succeed in controlling the infection, leading to an extinction of both intracellular and extracellular bacteria; or there will be an exponential explosion in the magnitude of the collective pathogenic population. Some interesting questions to explore may be: In the parameter space, which regions lead to eventual recovery, and which to persisting growth of infection? What shape do these regions take? And at the boundary between regions relating to stability and instability, is there a smooth transition, or are there any intricate structures? This last question being purely of analytical interest.

The first step will be to animate this model using the Processing framework similarly to as was seen in §4.

Appendix

A Demonstrations of boundedness

A.1 $f_L^{(1)}(t) < G(t) \forall t > 0$

$G(t)$, the probability of macrophage survival given initial infection with a single type b bacteria, and $f_L^{(1)}(t)$ a contrived lower bound of $G(t)$ are described as follows.

$$G(t) = \frac{G_+(1 - G_-) - G_-(1 - G_+)e^{u_0 t}}{(1 - G_-) - (1 - G_+)e^{u_0 t}} \quad f_L^{(1)}(t) = G_- + (1 - G_-)e^{u_0 t}$$

Remembering,

$$G_{\pm} = \frac{\eta}{2} \pm \sqrt{\frac{\eta^2}{4} - \frac{\mu_b}{\lambda_b}}, \quad \eta = \frac{\lambda_b + \mu_b + \delta}{\lambda_b}, \quad \& \quad u_0 = \lambda_b(G_+ - G_-).$$

$f_L^{(1)}(t)$ is chosen as a lower bound for $G(t)$ partly because of its similarity in behaviour –

$$\lim_{t \rightarrow \infty} f_L^{(1)}(t) = \lim_{t \rightarrow \infty} G(t) = G_-, \quad f_L^{(1)}(0) = G(0) = 1$$

– and also for the straightforwardness by which we can demonstrate that $f_L^{(1)}(t) < G(t) \forall t > 0$:

$$\begin{aligned}
f_L^{(1)}(t) &< G(t) \\
G_- + (1 - G_-)e^{u_0 t} &< \left(\frac{G_+(1 - G_-) - G_-(1 - G_+)e^{u_0 t}}{(1 - G_-) - (1 - G_+)e^{u_0 t}} \right) \\
(G_- + (1 - G_-)e^{u_0 t})((1 - G_-) - (1 - G_+)e^{u_0 t}) &< G_+(1 - G_-) - G_-(1 - G_+)e^{u_0 t} \\
G_-(1 - G_-) - G_-(1 - G_-)e^{u_0 t} + (1 - G_-)^2 e^{-u_0 t} - (1 - G_-)(1 - G_+) &< G_+(1 - G_-) - G_-(1 - G_+)e^{u_0 t} \\
G_-(1 - G_-) + (1 - G_-)^2 e^{-u_0 t} - (1 - G_+ - G_- + G_+G_-) &< G_+ - G_+G_- \\
-(1 - G_-)^2 + (1 - G_-)^2 e^{-u_0 t} &< 0 \\
e^{-u_0 t} &< 1
\end{aligned}$$

Which is in fact the case for all $t > 0$, provided $u_0 > 0$, this being true as long as λ_b, μ_b , & δ are not all 0.

A.2 $f_L^{(2)}(t) < S_0(t) \forall t > 0$

Strongly analogous to that seen above, the approximation for $S(t)$,

$$S_0(t) = \frac{S_+(1 - S_-) - S_-(1 - S_+)e^{v_0 t}}{(1 - S_-) - (1 - S_+)e^{v_0 t}}$$

is bounded from below by $f_L^{(2)}(t) = S_- + (1 - S_-)e^{v_0 t}$. Comonallities in behaviour between $S_0(t)$ & $f_L^{(2)}(t)$ align likewise to those in $G(t)$ & $f_L^{(1)}(t)$, and showing that $f_L^{(2)}(t) < S_0(t) \forall t > 0$ follows identical reasoning.

For reference:

$$S_{\pm} = \frac{\zeta}{2} \pm \sqrt{\frac{\zeta^2}{4} - [\mu_a - \gamma G_-]}, \quad \zeta = \frac{\lambda_a + \mu_a + \gamma + \delta}{\lambda_a}, \quad \& \quad v_0 = \lambda_a(S_+ - S_-).$$

B Plots representing bounded ranges

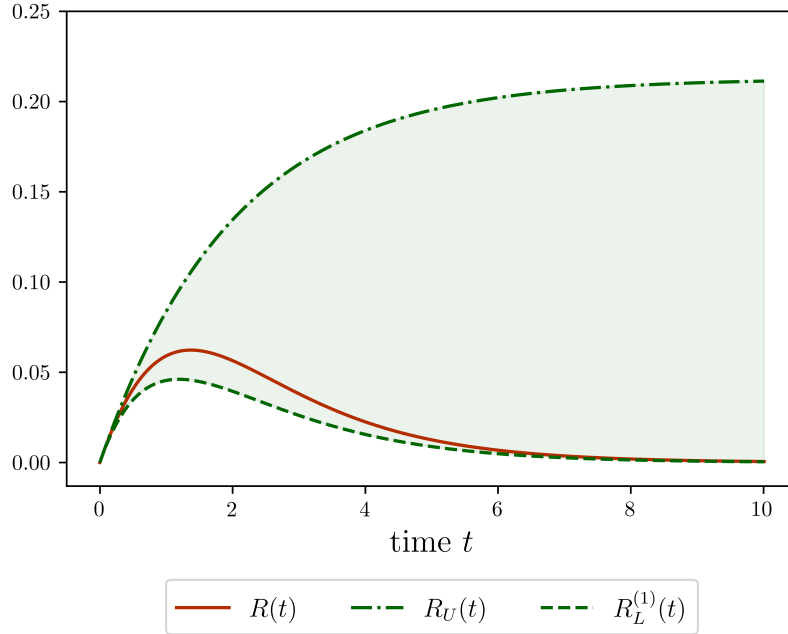


Figure 16: $R(t)$ stays between its upper and lower bounds, $R_U(t)$ & $R_L^{(1)}(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

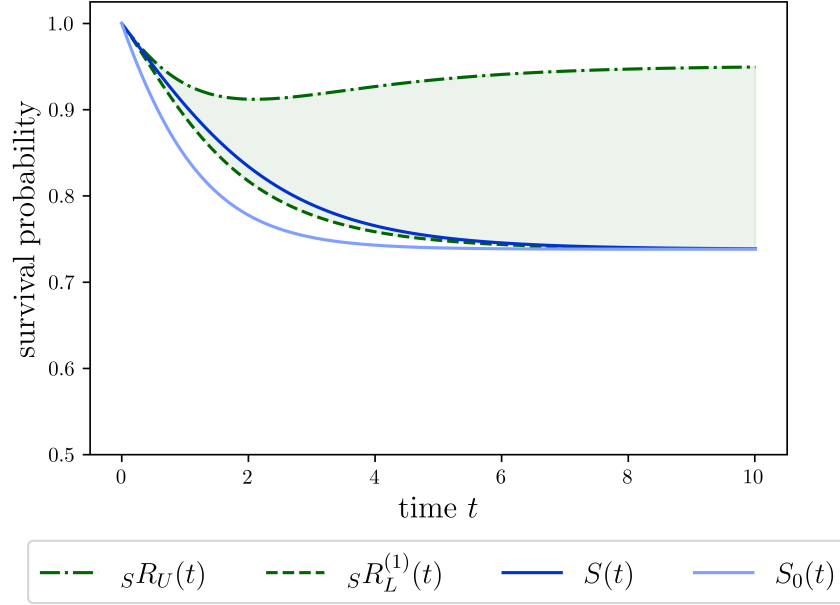


Figure 17: $S(t)$ stays between its upper and lower bounds that are implied by, $R_U(t)$ & $R_L^{(1)}(t)$, via the relation $R(t) = S(t) - S_0(t)$. Parameters used: $[\lambda_a, \mu_a, \lambda_b, \mu_b, \gamma, \delta] = [1.1, 1, 0.8, 1, 0.5, 0.1]$

C Open systems and the second law of thermodynamics

In addition to the mathematical modelling of inter-macrophage dynamics, my interest has been drawn to topics of a wider nature. Though the study of thermodynamics is indeed not directly related to immunology, it constitutes part of the background landscape that we as scientists traverse on our philosophical journey.

Prior to Carnot and Clausius who brought about a description of thermodynamic processes, the scientific understanding of nature rested in descriptions that had no fixture of a direction in time: Newton's equations of motion that can predict the trajectories of colliding spheres do not care whether the clock is moving clockwise or anticlockwise prior to the moment of physical interaction. We have come a long way since the birth of classical dynamics in our understanding of causality and predictability of various dynamics at play in nature [11]. It is now recognised that many systems are highly sensitive to initial conditions; and that there is often a level of complexity in causal interactions that surpasses the scope of rigorous description.

The latter of these challenges is nowhere more evident than in the study of biological processes [12]. Not only are the magnitude of causal relations beyond the scope of complete mathematical analysis; in many cases, they are yet even to be mapped out in entirety. But let us not lose hope. Much progress has been made – especially during the 20th century – towards getting a handle on this complexity. The probabilistic study of stochastic processes has come on in leaps and bounds in its efficacy towards this aim.

We have learned how to make highly informed best guesses as to how likely it is that this or that eventuality will come about – will a plague infection of a given initial bacterial load succeed in overthrowing the host's immune system, or will it be effectively repelled? And we may go further and ask: to what extent can we be sure of our prediction? These are the kinds of question I expect to continue asking as my career in science develops, but there is another simple question that I have been pondering on the side: What makes it all go?

What is the origin of the force that drives *Yersinia pestis* to continue the struggle for its proliferation; and what is it that compels the elaborate opposition from our immune systems against this assault?

The Darwinian view would suggest that the biological mechanisms which exhibit such teleonomy do so because it has been their predecessors that have been most successful in sticking around [13]. In the age old “fight for survival”, their ancestors (or more specifically, their genes [14]) have continually been effective in using the environment as a means of effective reproduction.

But in this so called “fight for survival”, from where exactly do the opponents get their gusto to keep going? Doesn't the second law of thermodynamics say that everything in the universe is on a one-way course to a mode of ultimate uniformity by way of heat death? This is indeed a property of systems that are isolated. But when there is a flow of energy or matter from an external environment, we can observe a seemingly different course of events, arising

from a potential that is characteristic of such a nonequilibrium situation.

An apparent contradiction

Why do far from equilibrium states tend to be order-producing? In an isolated system, it is stated by the second law of thermodynamics that entropy, S (disorder), will monotonically increase. Gas will spread to fill a space. Memory of its initial structural constitution is lost as the passage of time brings about a state of uniform equilibrium; multiple beginnings all proceed to the same end.

Why then in biological systems do we see such a plethora of emergent complex order, evolving into evermore differentiated forms?

It stands to be recognised that such systems are not in fact isolated – components interact with the domain superseding their boundary, and this occurs at multiple scales simultaneously: The bacteria draws from and secretes into its extracellular environment; the organ gives and takes substance, along with its concomitant neighbours, back and forth from the biological vehicle they are a part of; chimps interact with their communities, and more widely with the forest; trees absorb carbon dioxide from the air, thus tapping into the swirling collective of our global atmosphere. And still further out, the whole biosphere on earth is open to a constant flow of life-giving solar radiation, shooting at us through space at the speed of light from a star we call “the sun”.

Turbulent flow

Suppose you were to set up an experiment in the following way, similar to one proposed by Boltzmann. There are two chambers side by side. Initially, the right vessel is empty, and the left is filled with a large number of particles that behave as a gaseous fluid. Prior to running the dynamics, there is a seal blocking a narrow passage that connects these two spaces. This is made more important by the fact that, to begin with, the vessel containing a particulate substance is pressurised to a degree significantly greater than the one that is empty. A following detail is not vitally important to the experiment but is included for visually illustrative purposes. The left chamber filled with this substance – we’ll make it smoke for description – is further divided into two regions: upper and lower, such that the horizontal plane dividing the respective components of volume intersects exactly with the circular cross-section of the connecting channel, as a bisecting line. In addition to the passage being sealed at the experiment’s start, these sections of the left compartment, top and bottom, are filled with red and blue smoke respectively.

What happens when we open the seal, allowing this instantaneously contrived arrangement into a mode of freedom? You will point out that there will be a diffusion of colour between the upper and lower parts of the smoke-filled unit; a purple band will begin to grow at the boundary between red and blue. I would, however, like you to ignore this phenomenon – you may in fact assume there is a thin film dividing these two differently coloured regions – and instead, shift your attention to what will happen in the right compartment, and particularly to the space which is close to the aperture of the connecting channel.

Of course, the two colours of smoke will whoosh through the tube, burgeoning as a red, blue, and purple cloud in what was previously a void. As this process continues, more and more of the smoke will pass from the initially pressurised chamber to the other. As dictated by the second law of thermodynamics, this will continue until entropy is maximised; an equal mass of smoke will occupy each vessel. Also, at this final state of equilibrium, the red and blue will be completely mixed resulting in an effect where all smoke appears purple, but this detail is extraneous to the dynamic that I would now like to discuss.

Yes, as time proceeds in this experiment, entropy of the whole system, which can be considered to be closed, will monotonically increase; or as Boltzman would put it: the positioning of the particles will propagate towards a more probable occupation.

But what about if we consider the initially empty compartment independently, as an open system? How will the change in entropy, $\frac{dS}{dt}$, behave over time through the course of the experiment? The following quote is from Prigogine’s *Self-Organization in Nonequilibrium Systems* [15]:

“Let us consider the entropy change dS during a time interval dt . We may decompose it into the sum of two contributions:

$$dS = d_e S + d_i S$$

$d_e S$ is the entropy flux due to exchanges (of energy or matter) with the environment [(in this case the initially full vessel)] and $d_i S$ is due to the irreversible processes inside the system, such as diffusion, heat conduction, and chemical reactions.”

Indeed, the component of entropy change, $d_i S$, that captures only processes emerging within the initially empty vessel will be uniformly positive as the experiment proceeds. Something curious, however, will happen as the smoke first begins to travel through the aperture of the channel into the empty compartment. There will be a distinct generation of structure. Provided the Reynolds number of the fluid is sufficiently large, turbulence of delicately intricate and

ever more folding complex order will develop in and around the pluming jet of blue and red. Swirls will depart from whirls in a fractal descent, red and blue becoming interlaced in finer and finer layers.



In this localised portion of the system, there will be a marked reduction in entropy (in Boltzmann's terms: the number of particle complexions that correspond to the same form is small). When considering the right-hand compartment independently, it seems that this is confluent with – though a positive contribution to entropy change from internal processes ($d_i S > 0$) – a creative, order forming, chaotic dynamic inducing, negative entropy change from external influence ($d_e S < 0$). The second law of thermodynamics is never broken by this locally emergent order. And the reason for this is just that – the formation is “local”, and comes about during a global increase in entropy, as part of a wider diffusion from an initial condition harbouring latent potential.

Here we see an example whereby the resolution of a far from equilibrium situation results in an array of intricately ordered structures. It demonstrates how, as long as a system is open to some external environment for the transfer of energy or matter, order can come about with a corresponding reduction in entropy of the open system, and that this can happen without violating the second law of thermodynamics.

A wider view

Now the question is: can we draw a parallel here with the emergence of biological complexity? It seems we can observe the dynamics at multiple levels of analysis. The biosphere as a whole can be viewed as a large open system that receives a transfer of energy in the form of solar radiation from the sun. Whilst some of this energy is simply diffused into the earth's crust, and then partially radiated back into outer space, some of it is put to work. Quadrillions of leaves all around the world absorb these life-giving rays. By complex chemical mechanisms, this energy brings forth a creative dynamic whereby inert matter is brought to life in cellular structures. Then, from these initial streams of photosynthesis-driven order generation, there are secondary off-shoots of complex formation: leaf-eater ants traverse the rainforest vegetation, studiously making use of the abundant energy source locked up in these plant cells. Tertiary forms of life such as birds and small mammals take in a transfer of energy from the ants. The chain continues. Solar energy is diffused throughout the biosphere via a diverse multitude of varying interlacing pathways, passing to ever more and more differentiated and complex forms of life. It's like turbulence; an initial deviation, a fluctuation, in a flow that is latently far from equilibrium in nature, is amplified creating a growing perturbation. Then, on this first dysregulation towards a more differentiated state, a second layer of structure develops, on this, a third. An elaborate multi-faceted organisation spontaneously bursts into existence.

Can we line up an analogy to the fluid experiment? It may be a bit of a stretch, but it seems to me that the left-hand chamber full of smoke is like the sun emitting energy; the empty vessel is represented by the dead matter of the earth; and all the wondrously complex forms of biological life sprouting forth around the world are painted symbolically by the continual fractal descent of chaotic order in the turbulent whirls of red and blue smoke.

You may point out that life on earth has taken billions of years to develop to its current level of immense complexity, and that referring to this as a spontaneous bursting forth emergence seems a little off in the temporal sense. But who's to say 3 billion years is a long time?

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